

SANKHYA SCIENCE REFERENCE GUIDE v2.4

Quick Reference for INW Series Content Creators & Educators Document:

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Source Authority: All content derived exclusively from G. Srinivasan's authentic works:

- Secret of Sankhya: Acme of Axiomatic Unification, Part I (SoS1) — canonical edition 2021-04-21 (189 pp.), stored in-repo under `sources/` - Secret of Sankhya, Part II (SoS2) — canonical edition (571 pp.), stored in-repo under `sources/` - PHOSTATE.pdf, PHOscillator.pdf, PHO-GENETICS.pdf - SANKHYAABSTRACT.pdf, SANKEINSTEIN1.pdf - MEDITATION.pdf, SIDDHI.pdf

Audience: INW series writers, editors, educators — accessible for young adults (13-16)

Version History: | Version | Date | Changes | |-----|-----|-----| | 1.0 | June 2026 | Initial release | | 2.0 | June 24, 2026 | Category 7 analysis reports integrated; consolidated | | 2.1 | June 26, 2026 | 17 recommendations implemented (REC-3A-3F, REC-4A-4K); Parts 3 & 4 substantially expanded; new sections on Logic of Detection, PHO derivation chain, C-Power Ladder, hadronic/leptonic triads, Planck's constant derivation, field impedance, ferromagnetism, scale invariance | | 2.2 | June 27, 2026 | PHO Ratio = Gyromagnetic Ratio clarification; "Product Ratio" → "PHO Ratio" throughout | | 2.3 | June 28, 2026 | 14 technical corrections: §1.2 natural number counting; §2.2-2.3 terminology; §2.5 symbols; golden mean conjugate distinction throughout; §3.0a renumbered §3.1; Atho pre-polarized clarification; §4.5 kx/C^3 equation; §4.11 dark matter/energy rewrite incl. Hidden Two-Thirds paragraph; §10.4 Porous Pot analogy; §10.5 Mly count ladder; Changelog added | | 2.4 | July 16, 2026 | Aligned to canonical sources (SoS1 189 pp. + SoS2 571 pp., locked in-repo) and the new Master Symbol & Terminology Registry v1.0. Added §2.8a "The Two Light Speeds: C (Cs) vs c (CL)" reconciling the axiomatic count-rate with the measured light speed and documenting the SoS2 C-value inconsistency; added **Appendix A: Extended Derived Constants (Vetted)** with EM/atomic constants; §2.8 C row annotated as $C = C_s$; Registry cross-reference added throughout |

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PART 1: THE AXIOMATIC FOUNDATION

1.1 What Is Sankhya?

Sankhya (Sanskrit: counting, number, ratio, axiomatic reasoning) is an axiomatic science of counted interaction in a real, dynamic, coherent substratum. Created by Maharishi Kapila in a pre-Vedic period (~32,000 years ago), Sankhya forms the scientific core of the Bhagavadgita, embedded in the Mahabharata.

"It is the only unified field theory in existence that offers a complete scientific and mathematical solution to every aspect of reality, manifest or unmanifest, in space comprising substantial components." — G. Srinivasan, SoS1, Abstract/Overview

Critical distinction: Sankhya is spelled with the anuswara (ascending nasal sound), meaning "numbers, counting, reasoning." The common spelling "Samkhya" uses a descending labial sound that classifies it into a different category. We always use **Sankhya** per G. Srinivasan's authentic transliteration. (Source: SoS2, Preface, pp. 12–13)

1.2 The Core Axiom: 1 + 1 = 2

Everything in Sankhya derives from a single, provable truth:

Statement	Meaning
1 + 1 = 2	Natural number counting logic — the only axiom needed

This is the ONLY truth that: - Cannot be argued with - Needs no outside authority - Can be proven immediately - Contains within itself the seed of all complexity

The precise axiomatic significance of 1: The numeral "1" represents a real whole object — not an abstraction. Its properties are fully determined by counting logic alone:

- **1 is scale-invariant:** $1/1 = 2/2 = \infty/\infty = 1$. Any coherent whole, at any scale, counts as one. This is the foundation of Sankhya's scale-invariant law.
- **1/2 is the maximum change increment** — the largest single rate of change possible from a whole unit. Dividing the whole by 2 preserves a fixed center and produces the fundamental standing-wave ratio of all harmonic systems.
- **1/3, 1/4, 1/5, ... 1/n** represent progressively smaller change rates — decreasing incremental ratios as the denominator increases. The larger the denominator, the less change each increment represents.

This counting logic is why nature selects specific ratios, not arbitrary ones: only those that satisfy the balance equation $1/x = 1 + x$ are self-sustaining.

1.3 Criteria for a Correct Theory

G. Srinivasan establishes that a correct theory must be:

1. **Simple, logically & numerically consistent** and in consonance with reality
2. **Based on an axiomatic foundation** so causality is not violated
3. **Non-dimensional and scale-invariant** to fulfil universality
4. **Restricted to a single controlling variable** (time — the interval between activities)
5. **Self-evolving** — the definition of the substratum must evolve from within the theory
6. **Amenable to observation and practical verification**
7. **Proof generated from within** — not dependent on arbitrary external parameters

(Source: SoS1, Abstract/Overview; SANKHYAABSTRACT, p. 1)

1.4 The Sage Mindset vs. Modern Scientific Assumptions

G. Srinivasan identifies key intellectual errors in modern science and contrasts them with the Sankhya approach:

Modern Assumption	Sankhya Correction	Source
Space is a vacuum/void	Space is substantial, dynamic, coherent, holographic	SoS1, Overview; SoS2, Sutra 1
Constants are measured empirically	Constants are derived axiomatically from within	SoS1, Abstract
Uncertainty is fundamental	Counting interactions is certain and accurate	SoS1, pp. 8-9

Modern Assumption	Sankhya Correction	Source
Forces come in separate domains	There are no groups of forces — only different levels of simultaneous interactions	SoS2, Preface, p. 8
Perpetual motion violates energy conservation	PHO is nature's balanced condition in a substantial continuum	PHOSTATE.pdf
The universe is expanding	Axiomatically impossible in substantial space; Hubble's redshift has a different explanation	SoS1, Abstract; SANKHYAABSTRACT, p. 8
Growth patterns are observed, not explained	Patterns are derived as mathematically inevitable	SoS1, Axiomatic Derivations

"A classical example of a conundrum in Physics and Cosmology is the assumption that space is a vacuum, void or devoid of substantiality because of the failure of experiments to detect the presumed characteristics of space." — G. Srinivasan, SoS1, Abstract/Overview

"Since Universal phenomena existed long before the advent of human life, a correct theory must not depend on logic structured through intellectual analysis that is contrary to the observed characteristics of natural phenomena." — G. Srinivasan, SoS1, Overview

PART 2: THE MATHEMATICS OF DYNAMISM

2.1 The Balance Equation: $1/x = 1 + x$

At the transition point between collision, reversal, and separation in any interaction: -
 The **simultaneous state** (ratio within the instant) must equal a numerical ratio: $1/x$ -
 The **sequential state** (interval of separation) must equal the sum: $1 + x$

For balanced dynamism, they must be equal:

$$1/x = 1 + x \text{ — Equation [1.1]}$$

Solving with the axiomatic value $1 + 1 = 2$:

$$x = [\sqrt{(1 + 2^2)} - 1] / 2 = (\sqrt{5} - 1) / 2 = 0.618034 \text{ — Equation [1.2]}$$

Then: $1/x = 1 + x = 1.618034$ — Equation [1.3]

(Source: SoS1, Axiomatic Concepts, pp. 9-10; SANKHYAABSTRACT, pp. 4-5)

Why this number? It is the ONLY ratio that perfectly balances simultaneous activity ($1/x$) with sequential activity ($1 + x$). No other number satisfies this balance. Therefore, nature MUST use this ratio.

2.2 The First Slope of Interaction

SoS2 introduces the "first slope of an interaction." At the elemental level, the first meaningful relational difference is from one to two. The gradient can vary as 1:1 or 1:2. Taking the slope of 2 as a cyclic relational value:

Draw a right triangle with sides 1 and 2. The diagonal is $\sqrt{5}$. Remove the original unit 1, then share the remaining increment:

$$x = [\sqrt{(1 + 2^2)} - 1] / 2 = (\sqrt{5} - 1) / 2 = 0.618034 \text{ — Equation [1.2a]}$$

This is a visual derivation pathway: the first measurable relative change ratio between unity and duality. It is not merely geometry — it is the first visual signature of dynamism.

INW teaching phrase: "The first differential rate between one and two creates Nature's just-right dynamism share."

(Source: SoS2, First Slope of Interaction)

2.3 The $\sqrt{5}$ Connection Back to the Axiom

$$x = (\sqrt{5} - 1) / 2 = 0.618034, \text{ where } \sqrt{5} = \sqrt{(1^2 + 2^2)}$$

This connects the golden mean conjugate x directly back to the axiom $1 + 1 = 2$: $-1 = \text{unity} - 2 = \text{duality}$ (from $1 + 1$) - $\sqrt{5} = \text{the geometric bridge between unity and duality}$ - $x = (\sqrt{5} - 1) / 2$ — the golden mean conjugate; the primary cyclic variable in all Sankhya derivations - $\phi = 1/x = (1 + \sqrt{5}) / 2$ — the golden mean; derived from x as its reciprocal

In Sankhya's axiomatic system, **x is the primary variable** (the golden mean conjugate = 0.618034), and $\phi = 1/x$ is the derived reciprocal (the golden mean = 1.618034). The foundational balance equation $1/x = 1 + x$ is expressed directly in terms of x — ϕ appears only as $1/x$. This distinction matters: when G. Srinivasan writes equations using the variable x , he is working with the conjugate throughout.

x vs. ϕ — the key distinction: | Symbol | Value | Identity | Role in Sankhya |
|-----|-----|-----|-----| | x | 0.618034 | Golden mean conjugate = $1/\phi = \phi - 1$ | Primary cyclic variable in all derivations |
| ϕ | 1.618034 | Golden mean = $1/x = 1 + x$ | Expansion/reciprocal ratio; derived from x |

(Source: SoS1, Axiomatic Concepts)

2.4 The Self-Similar Loop

Transposing x : $1 = x + x^2$ — Equation [1.4]

Raising as powers (simultaneous ratios): $x^n = x^{n+1} + x^{n+2}$ — Equation [1.5]

The sum of infinite powers forms a loop: $\sum x^n = 1/x = 1 + x$ — Equation [1.6]

The combinatorial sum with Pascal triangle coefficients equals one: **Super-symmetric series** — Equation [1.7]

"The above sequence of derivational changes form an extraordinary and unique logical-loop, based on combinatorial values. It defines the characteristics of self-similarity and scale invariance." — G. Srinivasan, SANKHYAABSTRACT, p. 4

2.5 Disciplined Definitions: Self-Similar vs. Scale-Invariant

In G. Srinivasan's intended usage, **self-similar** and **scale-invariant** are disciplined technical terms — not casual synonyms. They are tied to Sankhya's logic of cyclic interaction-counts in a coherent substratum.

Term	Primary Meaning	Sankhyan Emphasis	INW Plain-Language
Self-similar	Same internal proportional rule	The process derives change from within itself	"The same balancing rule works inside the cycle."
Scale-invariant	Same rule across coherent levels	The rule remains valid when many synchronised interactions become a new unit	"When many become one, the same rule still works."

Self-similar interaction (disciplined definition): A self-similar interaction is one whose expansion, compression, and resonant exchange are all governed by the same internally generated proportional rule. The golden mean relation $1/x = 1 + x$ is self-similar because the same ratio repeats through powers of x and keeps cyclic exchange balanced without requiring an external scale or force.

- **Avoid reducing to:** "looks the same at different sizes" or "fractal-like"
- **Technical meaning:** internal proportional exchange in cyclic interaction

Scale-invariant law (disciplined definition): A scale-invariant law is a dimensionless interaction-count law that remains valid when the coherent unit changes scale. When many interactions synchronise and act as one, the same rule applies to that new unit.

- **Avoid reducing to:** "size does not matter"
- **More accurate:** "The same proportional law applies at each coherent level"

Swabhava (self-nature): The internally generated law of an interaction. In Srinivasan's usage, Swabhava is closely tied to self-similarity — the interaction derives its rule from its own internal condition rather than from an external agency.

Standing editorial rule: In INW, "self-similar" must always refer to an internally generated proportional rule of cyclic interaction. "Scale-invariant" must always refer to that same dimensionless rule applying when a coherent unit is redefined at another level. The golden mean equation $1/x = 1 + x$ is the anchor for both terms. **Do not use the terms interchangeably.**

(Source: SoS1, Axiomatic Concepts; SoS2, Self-Similar Derivations; Self-Similar Scale-Invariant Analysis)

2.6 Internal Exchange and Third-Order Damping

The golden mean conjugate ($x = 0.618034$) is an internally generated ratio. The unit exchanges from within itself around the central half:

Guna Mode	Operation	Expression
Thaama (compression)	Inward, density-forming	$x^2 = 1/2 - x^3/2$
Raja (resonance)	Reversal, balancing, damping	$x^3 = x - x^2$
Sathwa (expansion)	Outward, sequential, radiant	$x = 1/2 + x^3/2$

The x^3 term is the **third-order damping constraint** — it prevents collapse or runaway expansion and enables perpetual dynamism.

(Source: SoS2, Third Order Damping Constraint; Golden Mean Dynamism Insights)

2.7 The Ten-Count Cycle

Axiomatically, a resonant interactive cycle is limited to **ten interactions**:

- $\cos(2\pi/10) = 0.809017 = (1 + x)/2$
- A simultaneous cycle of 10 interactions = logarithmic index of 1
- All numerical values in Sankhya are presented as logarithmic indices to cyclic base ten

When sequential counting (1, 3, 6, 10, 15, 21, 28, 36, 45, **55**) meets simultaneous counting (0, 1, 1, 2, 3, 5, 8, 13, 21, 34, **55**, 89), they converge at the ten-count cycle at **55**. The next simultaneous ratio $89/55 \approx 1.618 = 1/x$.

(Source: SoS1, Axiomatic Concepts; SANKHYAABSTRACT, p. 5)

2.8 Key Axiomatic Constants

Constant	Value	Meaning	Source
x	0.618034	Golden mean conjugate — the cyclic interaction variable	SoS1, Eq. [1.2]
$1/x = \varphi$	1.618034	Golden mean — the expansion ratio	SoS1, Eq. [1.3]
$k = 2^{(1/3)}$	1.259921	Cube root of 2 — volumetric incremental ratio	SoS1, Eq. [1.19]
Kx	0.9149879	Purusha — maximum density boundary; equivalent to Catalan's constant	SoS1, Eq. [1.15]
$C = C_s$	296,575,967	Axiomatic count rate per cycle ($C_{\text{sankhya}} = 10^{(2/x^3)}$; the true field "speed of light." Distinct from the measured $c = CL$ — see §2.8a	SoS1, Eq. [1.16]
Mly (Moolaprakriti)	1.34462×10^{-51}	Minimum cyclic interval — smallest quantum unit	SANKHYAABSTRACT, p. 9
Mps (Mahad Prakriti)	2.20369×10^{-8}	Largest dynamic mass — equals Planck mass	SoS2, Symbols
PM (Prakriti Saptha)	1.67442×10^{-27}	Active nuclear spectrum — neutron/proton domain	SoS2, Symbols
Me (Mahad Vikrithi)	9.11023×10^{-31}	Reactive/observable unit — electron mass	SoS2, Symbols
Si	377.0376	Impedance to stress transmigration	SoS1, Eq. [1.20]

2.8a The Two Light Speeds: C (Cs) vs c (CL)

Sankhya distinguishes **two** "speeds of light," and keeping them separate resolves a long-standing point of confusion:

Symbol	Name	Value	What it is
C = Cs	Axiomatic count-rate (C_{sankhya})		The fundamental field interaction rate, derived purely from the axioms as C =

Symbol	Name	Value	What it is
		296,575,967 (2.9657596692×10^8)	$10^{(2/x^3)}$. Used in all Sankhya derivations.
c = CL	Measured light speed on Earth	299,792,458 (2.99792458×10^8)	The laboratory/terrestrial value physicists measure. It is the axiomatic rate blue-shifted by the correction factor Fc $\approx$ 1.010845 : CL = Cs × Fc .

Why they differ. The measured speed of light is not the bare field rate. Light propagating to a terrestrial observer is blue-shifted by the local gravitational/field gradient. In SoS1 the blue-shift factor is $F_c \approx 1.010845$ (its logarithm equals R_s/R_o), so the measured value $CL = C_s \times F_c \approx 299,792,458$ — exactly the CODATA figure. The axiomatic count-rate C_s is the physically primary quantity; CL is what an instrument on Earth reads.

Convention (per Registry §2.2). Uppercase **C = Cs** is the axiomatic count-rate; lowercase **c = CL** is the measured terrestrial speed of light. Always use **C = 296,575,967** in derivations.

Note on the SoS2 inconsistency. SoS2 is not internally consistent on the value of C. Its Symbols/Definitions table and some passages quote **C = 2.99792459×10^8** (the measured value), whereas every axiomatic derivation in the same volume uses **C = 296,575,967**. The discrepancy is visible across **SoS2 pp. 198, 295, 514 and 567**. This is an editorial inconsistency, **not** two different constants: read the Symbols-table figure as $c = CL$ (measured) and the derivation figure as $C = C_s$ (axiomatic). The Master Symbol & Terminology Registry v1.0 (§2.2) records this resolution.

2.9 The Fibonacci Sequence

Definition: Each number is the sum of the two preceding numbers.

$$F(n) = F(n-1) + F(n-2)$$

Position	Value	Ratio $F(n+1)/F(n)$
F(8)	21	1.615
F(9)	34	1.619
F(10)	55	1.618
F(11)	89	1.618

Position	Value	Ratio $F(n+1)/F(n)$
F(12)	144	1.618

As n increases, $F(n+1)/F(n) \rightarrow \phi$. The Fibonacci sequence is the discrete manifestation of the continuous golden ratio.

PART 3: THE SUBSTRATUM AND ITS PROPERTIES

3.0 Why Sutra 1 Is the Master-Key

The very first Sutra of the Sankhya Karika is not a simple opening line. It is the compressed seed of the entire 68-sutra system — the core theorem from which all else follows.

"The very first Suthra in almost all important Sanskrit works contains the core theorem." — G. Srinivasan, SoS2, p. 141

"Suthra 1 is a pivotal theorem and provides the basis to understand subsequent Suthras." — G. Srinivasan, SoS2, p. 142

G. Srinivasan devotes significant space in SoS2 to establishing that Sutra 1 already contains "the essential core mathematical principles in brief from all the Suthras." The explanation of this single sutra spans pp. 141–176, covering the definition of the substratum, the three stress modes, the four required field conditions, and the logical chain from detection to holographic manifestation.

Structural closure: Sutra 68, the final theorem, closes the system by repeating the key terms "aikanthikam-aathyanthikam" — confirming that the entire treatise is a self-consistent logical loop returning to its starting point. (Source: SoS2, p. 44, Tribute to Kapila)

INW teaching framing: "The first sutra states the model; the remaining 67 sutras unpack it."

Four questions for students: 1. What is seen? → Vibrations and changes 2. What is not directly seen? → The balanced background substratum 3. What creates the visible world? → Three recurring modes of interaction (stress) 4. Why does the first sutra matter so much? → Because it names the hidden base, the visible effects, and the rule that connects them

(Source: SoS2, pp. 141–142; Sutra 1 Report)

3.1 The Detection Principle

Before examining the substratum's properties, a foundational axiom must be stated explicitly:

Only a change can be detected.

"Only a change can be detected and an interaction between components or objects causes a change that can be observed." — G. Srinivasan, SoS2, p. 143

An interaction that IS observable causes a detectable vibration. An interaction that is NOT observable causes a **stress** — a hidden force or pressure that remains real but undetected.

This is the reason non-detection does not mean non-existence. If the fundamental components of space are in perfectly balanced interaction, they produce no net change to detect — yet they remain real, substantial, and active.

INW teaching point: "You can only see what changes. If something is perfectly balanced, it's invisible — but it doesn't stop being there."

(Source: SoS2, p. 143)

3.2 The Four Properties of the Substratum (Sutra 1)

The very first Sutra defines the substratum by four axiomatic properties. In Sanskrit: "Aikaantha-Athyantha-Atho-Abhavath"

Sanskrit	Meaning	Physical Implication
Aikaantha	Coherent, synchronised, solid	Many interactions acting as one unit
Athyantha	Perpetual, endless, omnipresent	The field never decays — eternally dynamic
Atho	Dynamic, oscillating, moving	Continuously vibrating and interacting
Abhavath	Unmanifest, undetectable, quiescent	In balanced state, both intervals equal — hence undetectable

"The most important axiomatic definition of space in the first Suthra, 'Aikaantha-Athyantha-Atho-Abhavath' (Coherent, Perpetual, Dynamic but Unmanifest) as the substratum of phenomenon was overlooked completely by all the previous authors." — G. Srinivasan, SoS2, Preface, pp. 12-13

(Source: SoS2, Preface pp. 12-13; confirmed by repetition in Sutra 68)

3.2.1 Aikaantha — Coherence and Synchronisation

Aikaantha describes the coherent or synchronised state of the substratum. It is active internally but outwardly stable and hard to detect because its internal exchanges balance perfectly.

Key characteristics: - Multiple interactions act as a single unified entity - Internal activity is intense but self-cancelling from the outside - This is why the substratum appears "solid" or "still" — not because nothing is happening, but because everything is happening in perfect unison - Connects directly to the "vanishing counts" principle (see §3.4): when many claps occur simultaneously, they merge into one — nine counts vanish into hidden density

Physical analogy: An orchestra playing in perfect unison produces one coherent sound, not a hundred separate notes. The individual instruments are all active, but their synchronisation makes them function as one.

Connection to Sutra 68: The closing theorem repeats "aikanthikam" to confirm that coherence is both the starting and ending condition of the treatise.

(Source: SoS2, Sutra 1 analysis, pp. 161-169)

3.2.2 Aathyantha — Perpetual Continuity

Aathyantha establishes that the substratum's cyclic activity is perpetual. It does not simply run down, decay, or stop. This is not an assumption imposed from outside — it follows axiomatically from the balanced nature of the substratum itself.

Key characteristics: - The field's oscillation is self-sustaining because reactive displacement never exceeds interactive displacement - No energy "leaks out" — the balanced exchange is a closed, self-renewing cycle - This property is the foundation for the PHO state (forward-link to Part 4) - Perpetuity is a mathematical consequence of the golden mean balance, not a philosophical assertion

INW teaching point: "The substratum doesn't run down like a battery. Its balanced beat IS what 'being' means — it has no reason to stop, because stopping would require an external cause, and nothing is outside it."

(Source: SoS2, Sutra 1 analysis, pp. 170-171)

3.2.3 Atho — Pre-Polarized Latent State

Atho is the pre-manifestation, pre-polarization, pre-radiation state of the substratum. It is not yet differentiated into Sathwa/Raja/Thaama — it exists as an undifferentiated, latent potential condition that precedes all observable interaction.

Key characteristics: - Atho is a **pre-polarized state** — it has not yet differentiated into the three Guna modes - It is **latent, potential, undifferentiated** — containing the seed of manifestation without yet manifesting - Without Atho, the field would remain perfectly hidden forever — no universe could manifest - The Atho state is not a flaw or defect — it is the indispensable pre-radiated condition through which the unmanifest becomes manifest - Atho is the **undifferentiated potential** that, when conditions are right, tips into polarization and allows the universe to become visible

INW teaching point: "If the substratum had no latent pre-polarized potential at all, nothing would ever appear. Atho is the undifferentiated state before manifestation — the field's dormant readiness, not yet a force but ready to become one."

(Source: SoS2, Sutra 1 analysis, pp. 171-172)

3.2.4 Abhaavath — Balanced Unmanifestation

Abhaavath describes the condition in which all internal exchanges remain perfectly balanced, producing zero net displacement. In this state, the substratum is unmanifest — real but undetectable.

Key characteristics: - When simultaneous and sequential intervals are equal, there is no difference to detect - This explains directly why the Michelson-Morley experiment and all similar experiments failed to detect space as "aether" - Non-detection is the expected result of a perfectly balanced substratum, not evidence of non-existence - Abhaavath is the complementary condition to Atho: where Atho opens manifestation, Abhaavath closes it back into hiddenness

The detection chain: - Balanced interaction (Abhaavath) = zero difference in count - → prevents stress-transmigratory states - → therefore prevents radiant/manifest states

INW teaching point: "The field doesn't hide because it's shy. It hides because it's so perfectly balanced that there's nothing to see. Like a spinning top balanced on its point — all the action is there, but from outside it looks like nothing is happening."

The 2×3 Guna matrix in the Atho state: In the 2×3 matrix of Guna interactions describing the Atho (pre-polarized) state, the Sathwa component equals zero. Only Thaama and Raja components are present — compressive density and resonant shuttling — but no expansive/radiant mode. The required balance condition is:

$$(\pi \times x) / e = (7 \times RS) / 10$$

where $\pi = 3.14159...$, $x = 0.618034$, $e = 2.71828...$, $RS = 1.020408$. In this state there is no observable mass, charge, or radiation — only pre-polarized, latent stress activity that has not yet differentiated into manifest form.

(Source: SoS2, Sutra 1 analysis, pp. 173-174)

3.3 Why Space Cannot Be Detected

In a balanced state, the simultaneous and sequential intervals remain equal. There is no difference to detect. This is why Michelson, Morley and others failed totally to detect space (as aether) — the substratum is self-cancelling in its balanced state.

"In a balanced state both intervals remain equal. Hence space must remain undetectable. It is the main reason why all efforts to detect space (as aether) failed totally." — G. Srinivasan, SANKHYAABSTRACT, p. 9

Key INW teaching: Non-detection is not non-existence. Invisible is not unreal.

3.4 Simultaneity and Sequence

These are the two fundamental modes of activity in the substratum:

Simultaneity: - All components act together within one instant - Many interactions merge into a single count - Creates density, mass, coherence, potential - Synonymous with: mass, density, rigidity, coherence

Sequence: - Components act one-after-another over time - An interval separates two adjacent activities - Creates detectable change, displacement, motion

The Vanishing Counts:

"Ten claps can be counted as such if it occurs one after another but if all ten occur at the same instant it can be counted as only one. It causes an anomaly of nine vanishing counts when interactions synchronise to act simultaneously." — G. Srinivasan, SoS1, pp. 8-9

This "clap analogy" is a key teaching tool for INW.

3.5 The Three Phases of Interaction

Any cycle of interaction between two objects has three complementary phases:

Phase	Time Mode	Stress Type	Guna
Compressive collision	Simultaneous instant	Compressive stresses at impact	Thaama
Rebounding reversal	Resonant period	Compressive reverses to expansive	Raja
Expansive separation	Sequential interval	Stress transmigrates beyond boundary	Sathwa

(Source: SoS1, Overview; SANKHYAABSTRACT, pp. 2-3)

3.5.1 The Guna Equation S.1 — Interactive Count Formulae

G. Srinivasan provides the mathematical expression for the three Guna modes as interactive count formulae in the Sutra 1 analysis itself:

Guna	Interactive Count Formula	Physical Mode	Domain
Thaama	$C^{(1+x)}$	Compressive, dense, coherent	Hadronic/nuclear — strong interaction domain
Raja	$C^{(x+x)} = C^{(2x)}$	Resonant, shuttling, bonding	Interface — weak interaction domain
Sathwa	$C^{(1-x)}$	Expansive, radiant, elastic	Leptonic/electromagnetic — radiation domain

Where: $C = 2.9657596692 \times 10^8$ (axiomatic oscillator rate); $x = 0.618034$ (golden mean conjugate)

Verification: $C^{(1+x)} \times C^{(1-x)} = C^2$ — the product of Thaama and Sathwa gives the full two-axis count $C^{(2x)} = C^{(1.236068)}$ — the Raja/resonant mode sits between the compressive and expansive modes - The three formulae together describe the complete interactive spectrum from hidden density to radiant output

INW teaching point: "The three Gunas aren't just descriptions — they're precise mathematical ratios of the same field oscillator C , operating in three different modes simultaneously."

(Source: SoS2, p. 145, Equation S.1)

3.6 Space Is Not Empty — Key Arguments

G. Srinivasan provides several axiomatic arguments:

1. **It is axiomatically impossible** to accept that the foundation of the most realistic Universe is a void

2. A real component or elemental matter must be **volumetric** — its dimensional property of length defines the unit value
3. Since there can be no smaller unit in an elementary field, the volume is **unit cubed**
4. The next incremental dimension must be 2 units, giving **$2^3 = 8$ volumes** (1 original + 7 incremental)
5. A fractional dimension is **axiomatically forbidden** in an elemental field
6. Hence the quantum state of 1 containing 7 unit volumes is a **natural axiomatic consequence**

(Source: SoS1, Overview, pp. 6–7)

3.7 The Logic of Detection

G. Srinivasan presents a step-by-step logical chain (SoS2, pp. 158–159, 174–176) that demonstrates how the nature of the substratum can be deduced from the simple premise that only change can be detected. This chain is both rigorous and teachable:

Step 1 — Foundational premise: Only change can be detected.

Step 2 — Observational premise: The observer detects only vibrations. Therefore, all manifested phenomena are aggregates or collective states of vibrations on a fundamental base.

"Only vibrations are detected by the observer and all manifested phenomenon are aggregates or collective states of such vibrations on a fundamental base." — G. Srinivasan, SoS2, p. 174

Step 3 — Causal premise: Vibrations are actions. Therefore, they must be caused by fundamental objects or components.

Step 4 — Epistemic limit: Only relative changes can be detected. Therefore, non-vibratory fundamental components cannot be detected directly.

Step 5 — Mechanism: Vibrations arise from components in interaction. Interaction = cyclic colliding and separating.

Step 6 — Three stress modes: From the interaction of components, three stress modes are produced: - **Collision** → compressive / inelastic stresses - **Reaction** → expansive / elastic stresses - **No freedom to move away** → cyclic / shuttling / resonant stresses

Step 7 — Measurement consequence: - If there is oscillatory displacement (to-and-fro movement) → vibration can be measured - If there is no space to move → vibratory action

becomes stresses/pressures → these stresses transmigrate across components in the three modes

Step 8 — Ontological conclusion: If detection depends on vibrations → fundamental space must contain components with the four characteristics/states (Aikaantha, Aathyantha, Atho, Abhaavath) → therefore space functions holographically → all vibrations remain in fixed relationship relative to each vibratory point → in a coherent and resonant state.

"The following Suthras provide logically concatenating proof to show that manifestation is a holographic phenomenon." — G. Srinivasan, SoS2, p. 159

Clarification note on "holographic": G. Srinivasan's use of "holographic" means that all vibrations remain in fixed positional relationships relative to each other in a coherent/resonant state. This is a detection-theoretic concept rooted in Sankhya axiomatics — it describes a property of substantial space where each point contains the relational pattern of the whole. It is not identical to the "holographic principle" in modern theoretical physics (which concerns boundary/bulk information encoding).

INW teaching summary: "Start from one simple truth: you can only see what changes. Follow the logic step by step, and you arrive at the conclusion that space must be substantial, coherent, dynamic, and holographic. That's what Sutra 1 tells us in compressed form."

(Source: SoS2, pp. 158-159, 174-176; Sutra 1 Report)

3.8 The Holographic Function of Space

The Logic of Detection concludes that space functions holographically. This has specific implications:

1. **Every point contains the pattern:** In a coherent, resonant field, every vibratory point maintains a fixed relationship to every other point. The pattern is not localised — it is distributed throughout the field.
2. **Detection is relational, not absolute:** We do not detect the substratum directly; we detect the vibrations (changes) within it. These vibrations form fixed relational patterns — what G. Srinivasan calls the holographic form.
3. **The source of the holographic state:** Both the $1/2$, x , and x^2 locations act simultaneously within the interaction cycle, forming the source of a holographic state.

"What Maharishi Kapila had demonstrated through axiomatic mathematics was that all phenomena in nature was of a holographic or coherent vibratory form." — G. Srinivasan, SoS2, Preface, p. 13

(Source: SoS1, Axiomatic Concepts; SANKHYAABSTRACT, p. 5; SoS2, p. 159)

3.9 Scale Invariance of the Substratum

G. Srinivasan explicitly states that the same axiomatic laws apply at every scale of manifestation, from subatomic particles to galaxies:

"The same laws apply in evaluating the galaxy, sun, planets, protons, electrons etc. The distance always varies with cyclic interactive period keeping C the perpetual oscillatory rate constant." — G. Srinivasan, SoS2, p. 159

Key implications: - The laws derived from the axiomatic foundation are not limited to any particular scale - C (the oscillator rate) is the constant; distance and period vary - This is why Sankhya can derive values for both Planck-scale quantities and nuclear/atomic quantities from the same axioms - Scale invariance is not merely an observation — it is an axiomatic necessity in a coherent, substantial field

INW teaching point: "The same rules that explain how a proton stays together also explain how a galaxy holds its shape. In Sankhya, there is only one law — it just shows up at different levels."

(Source: SoS2, p. 159)

PART 4: THE PHO STATE — ENGINE OF THE SUBSTRATUM

4.1 What Is the PHO State?

PHO = Perpetual Harmonic Oscillator — the ubiquitous engine of the substratum. It is the self-maintaining cyclic condition through which all manifestation becomes intelligible.

PHO Element	Meaning	Manifest Expression
Perpetual	Does not decay — reactive displacement never exceeds interactive displacement	Persistence of substratum and stable matter
Harmonic	Exact ratio, phase, standing-wave, self-similar count	Stable spectra, mass ratios, resonant states
Oscillator	Cyclic expansion, compression, resonant exchange	Wave, particle, charge, mass as phase conditions
Volumetric	Spherical/volumetric — manifests density, boundary, flux	Hadronic cores, leptonic surfaces, charge-like flux
Simultaneous	Many interactions within one cycle and one common center	Hidden counts become mass, density, potential

"PERPETUAL HARMONIC OSCILLATION IS AN AXIOMATIC STATE OF SPACE." — G. Srinivasan, PHOSTATE

(Source: PHOSTATE.pdf; PHOscillator.pdf; Key Insights Report)

4.2 The Bow Analogy (PHO Physical Model)

"An archer's bow is the iconic model of the PHO state. The shaft of the bow is a flat long strip made up of a rigid material like wood or metal. When it is bent the outer curve stretches while the inner surface compresses at the same time... If a string is attached to both the ends when bent, it would resonate or vibrate... Depending on the rigidity of the strip and the elasticity of the string it would reach a balanced resonant state and remain in that state forever." — G. Srinivasan, SoS1, Overview; SoS2, Preface

An amplifier with a microphone attached confirmed this perpetual resonance experimentally.

4.3 From Axioms to the PHO: The Derivation Chain

Part 4 of the v2.0 Guide presented PHO ratios as a table of values without showing how they are derived from the axiomatic foundation. The following derivation chain, drawn from SoS2's "Power House of the Universe" chapter (pp. 353-362), traces the complete path from axiom to PHO:

Step	Operation	Result	Significance
EP1	Start: $1 + 1 = 2$; standing-wave ratio $1/2$	$2^3 = 8$ volumes (7 incremental)	Spectral states sum to $1/7$; quantum structure emerges
EP2	$k = 2^{(1/3)} = 1.259921$	Radial-to-volumetric ratio	Volume doubles when radius increases by $k-1$
EP3	Two simultaneous 10-count cycles / sequential volumetric displacement	$PR = 100/28 = 3.571428$	PHO Ratio (= Gyromagnetic Ratio) — the perpetual oscillatory ratio
EP4	Asymptotic sum of combinatorial simultaneous interactions	$Kx = 0.91499$	Purusha state — maximum coherent delay/inertia/mass
EP5	Self-similar harmonic ratio as cyclic count	$C = 2.9657596692 \times 10^8$	Fundamental oscillator rate of the field
EP6	Kx / C^6	$My = 1.3446 \times 10^{-51}$	Moolaprakriti — minimum interactive unit

"The ratio EP5 is a simultaneous self similar ratio and as it's a cubic ratio of x it has the limiting property or damping force to maintain an oscillatory state perpetually."
— G. Srinivasan, SoS2, p. 356

Teaching sequence for INW: 1. Start with the axiom: $1 + 1 = 2$ 2. Divide by 2 — this preserves a fixed center and generates stable nodes 3. From that geometry, derive the volumetric quantum (8 volumes = 7 incremental + 1 original) 4. Build the resonant ratios (PR and RS) from interaction counts 5. Derive C (the oscillator rate) from the self-similar properties 6. Derive the full range: maximum (Kx) to minimum ($My = Kx/C^6$) 7. Show that stable particles and constants are states on this ladder — not isolated objects in empty space

(Source: SoS2, pp. 353–357, derivations EP1 through EP6)

4.4 Key PHO Ratios

Ratio	Value	Meaning	Derivation
Standing-wave ratio	$1/2$	Fundamental balanced division	Division by 2 keeps center fixed
$k = 2^{(1/3)}$	1.259921	Radial-to-volumetric increase ratio	Cube root of 2
	$2^3 = 8$		—

Ratio	Value	Meaning	Derivation
Volumetric doubling		A radius doubling gives 8 volumes (7 incremental beyond original)	
PR (PHO Ratio)	$100/28 = 3.571428$	Two simultaneous 10-count cycles (100) divided by sequential forward/reverse volumetric displacement (28). PR = Gyromagnetic Ratio (Gmr).	EP3
RS (Resonant Ratio)	$100/98 = 1.020408$	Resonant restoring factor; also $PR \times (2/7)$	EP3 derivative

4.4.1 RS as Transcendental Power Series

The Resonant Ratio RS is not merely a numerical value — it is derivable as a transcendental power series:

$$RS = \Sigma(2/100)^n \text{ for } n = 0 \text{ to } \infty = 100/(100 - 2) = 1.020408...$$

This is the "rate of decay of interactions per cycle" as a perpetually dynamic state. The key insight: the coherent state's decay time, when expressed through this series, is infinite/eternal. This is why the PHO is perpetual — the restoring factor ensures the oscillation never dies out.

Physical basis of RS = PR × 2/7: - 2 = two changes of volume per cycle (expansion and compression) - 7 = seven volumetric displacements within the quantum

(Source: PHO STATE; PHO Oscillator)

4.4.2 PR as Nuclear Gyromagnetic Ratio

"THE PR = 3.5714 VALUE IS ALSO THE OBSERVED NUCLEAR GYROMAGNETIC RATIO BUT HAS NOT BEEN THEORETICALLY DERIVED SO FAR IN PHYSICS." — G. Srinivasan, PHO STATE

The PHO Ratio $PR = 100/28 = 3.571428$ is axiomatically derived from Sankhya's interaction counts. This ratio is identical to the Gyromagnetic Ratio (Gmr) — the measured nuclear gyromagnetic ratio in experimental physics — which has not been theoretically derived by modern physics. G. Srinivasan presents this identity ($PR = Gmr$) as confirmatory evidence of the PHO framework: Sankhya axiomatically derives a value that physics can only measure.

(Source: PHO STATE)

4.5 The Ladder of Phenomenon: C-Power States

The C-power ladder is the central organising structure of the PHO chapter and the key to understanding how the substratum field produces different states of manifestation. It is built from powers of C (the fundamental oscillator rate):

Power	State	Physical Domain	Significance
C	Fundamental oscillator rate	Elemental field cycle	$\approx 2.9658 \times 10^8$ counts/cycle
C²	"Vanishing" from detectable spectrum	Synchronised axis counts	Two-axis synchrony; counts become hidden
C³	Hadronic / coherent domain	Nuclear stable state	Non-radiant; mass/density; 3D coherent count where all three axes synchronise
C⁴	Intermediate transition	Transition domain	—
C⁵	Planck-mass (Mahad) level	Maximum oscillatory state	Connects to Planck scale; operates at C cycles/cycle
C⁶	Maximum field count	Two C ³ states interacting simultaneously	Upper coherent limit

Key teaching points: - **C³ is the stable 3D coherent count** where all three spatial axes synchronise. This creates the hadronic (nuclear) domain — dense, non-radiant, stable matter. - **C⁶ is the maximum** when two such C³ states interact simultaneously. This defines the upper coherent limit of the field. - **K_x (maximum) and M_y = K_x/C⁶ (minimum)** define the full dynamic range of the substratum — from the densest coherent state to the smallest quantum unit. - Stable particles and constants are not isolated objects — they are **states or transitions on this ladder**.

INW teaching analogy: "Think of the C-power ladder as a building with floors. The ground floor (C) is the basic beat. Each higher floor packs more beats together simultaneously. By the time you reach the top floor (C⁶), so many beats are packed together that nothing can escape — that's the deepest hidden potential. Going down, the beats spread out and become visible as particles, waves, and radiation."

(Source: SoS2, pp. 361–362; SoS2, p. 368, Fig: The Ladder Of Phenomenon)

Connection between K_x and leptonic mass — the C³ symmetry:

From the foundational derivation $K_x = M_y \times C^6$, a key symmetry equation follows directly:

$$Kx / C^3 = Mly \times C^3$$

Equivalently: $Kx = Mly \times C^6$

This equation reveals that the maximum coherent boundary (Kx, the Purusha constant ≈ 0.9149879) and the minimum quantum unit ($Mly \approx 1.3446 \times 10^{-51}$) are related through the C^3 factor — exactly the hadronic/leptonic boundary level on the Ladder of Phenomenon. The C^3 level is where all three spatial axes synchronise to form stable hadronic (nuclear) states. The equation $Kx / C^3 = Mly \times C^3$ shows that both ends of the Ladder — the maximum and minimum — are symmetrically anchored at the C^3 (hadronic/leptonic) boundary, connecting the Kx constant directly to the domain of leptonic mass.

(Source: SoS1, Eq. [EP6]; SANKHYAABSTRACT, p. 9)

4.6 Hadronic States: Proton, Neutron, and the Hidden Coherent Mass

The three hadronic states form the "nuclear triad" — three precisely derived masses that demonstrate the PHO state at the nuclear level. Their perpetual oscillatory relationship via PR and RS is the core demonstration of the PHO in action.

State	Symbol	Axiomatically Derived Value	Physics Measured Value	Match Error
Neutron	Pn	$1.6749276458 \times 10^{-27}$	$1.67492728 \times 10^{-27}$	$+2.18 \times 10^{-7}$
Coherent mass	PM	$1.6744231791 \times 10^{-27}$	Not known in physics	—
Proton	Pm	$1.672621512 \times 10^{-27}$	$1.67262171 \times 10^{-27}$	$+1.18 \times 10^{-7}$

The PHO proof — perpetual oscillatory ratios maintained:

$$(PM - Pm) / (Pn - PM) = PR = 3.5714285714$$

The proton mass change occurs over 7 volumetric displacements; the neutron mass change occurs over 2 volumetric displacements within the same oscillatory period.

$$(PM - Pm) / (Pn - PM) \times (2/7) = RS = 1.0204081633$$

The hidden coherent mass PM: This is a state that modern physics has not identified. In Sankhya, it is the balanced, coherent center between the proton and neutron — the stable reference point around which both oscillate perpetually.

"The reader must realise that at the atomic / nuclear level there is no uncertainty what so ever." — G. Srinivasan, PHOSTATE

INW teaching point: "The proton and neutron aren't just sitting there — they're perpetually oscillating around a hidden balanced center. The ratios between them are exact, not approximate. There is no uncertainty at this level — only precise, axiomatic counting."

(Source: PHOSTATE; PHOscillator; PHO-GENETICS — all three confirm identical derivations)

4.7 Leptonic States: Electron as Boundary, Not Independent Particle

The leptonic states demonstrate that the electron is NOT an independent particle — it is the harmonic oscillatory boundary of the hadronic nuclear core. This is a paradigm-shifting claim backed by precise derivation.

State	Symbol	Derived Value	Physics Measured	Notes
PM-derived lepton	Mep	$9.1140580241 \times 10^{-31}$	Not predicted	—
Planck-mass-derived lepton	Me	$9.110233722 \times 10^{-31}$	Not predicted	—
Balancing state (= measured electron)	Mee	$9.1093838239 \times 10^{-31}$	$9.1093897 \times 10^{-31}$	Extremely precise match

The PHO proof at the leptonic level:

$$(\text{Mep} - \text{Me}) / (\text{Me} - \text{Mee}) \times (\text{k}^2/7) = \text{RS} = 1.0204081633$$

The same resonant ratio RS that governs the hadronic triad also governs the leptonic triad — demonstrating scale invariance across the nuclear and electronic domains.

Key teaching points: - Leptonic states are **area-dependent flux densities** displaying "charge" characteristics - They act as a **simultaneous surface of interaction** at distance k from the hadronic center - The electron (Mee) is the **resonant balancing state** of the boundary, not a free-floating independent particle - The leptonic triad uses k^2 (area ratio) rather than k^3 (volume ratio) because it describes a surface, not a volume

"THE LEPTONIC STATES ARE ONLY THE BOUNDARY STATE OF THE HADRONIC NUCLEAR CORE STATE. THEREFORE THE LEPTONIC ELECTRON IS NOT AN INDEPENDANT PARTICLE." — G. Srinivasan, PHOSTATE

INW teaching point: "The electron isn't a little ball orbiting the nucleus. It's the vibrating boundary of the nucleus itself — like the surface of a drum isn't separate from the drum. The same exact ratio (RS) that governs the proton-neutron relationship also governs the electron relationship."

(Source: PHOSTATE; PHOscillator; PHO-GENETICS)

4.8 Planck Mass (Mahad) as PHO Oscillator

The Planck mass, identified in Sankhya as Mahad Prakriti (Mps), is not merely a theoretical threshold. It is a dynamic, perpetual, self-similar oscillator operating at exactly C cycles per cycle.

"A dynamic perpetual self similar oscillator operating at exactly C cycles per cycle (or second)." — G. Srinivasan, SoS2, pp. 358-359

Key characteristics: - The Mahad (Planck mass) sits at the C^5 level on the Ladder of Phenomenon - It represents the interface between the Thaama/Raja domains (strong/weak force interface in physics) — the EP3 ratio equivalence governs this transition - The transition from Planck mass to neutron **does not radiate any energy** — hence it is not detectable until a violation of the coherent state occurs - The Compton wavelength exists at the boundary $k-1 = 0.2599$ from the coherent surface

"This transition does not radiate any energy. Hence it is not detectable till a violation of the coherent state occurs." — G. Srinivasan, SoS2, p. 359

INW teaching point: "The Planck mass isn't just a number — it's an active oscillator, beating at exactly C beats per second. It's the highest-energy stable oscillator in the field ladder, and it connects the quantum world to the nuclear world."

(Source: SoS2, pp. 358-360)

4.9 Planck's Constant: Axiomatically Derived

G. Srinivasan describes the axiomatic derivation of Planck's constant from the PM (coherent nuclear mass) state as "a unique and unequivocal proof of Sankhyan concepts and logic." This derivation demonstrates unification across quantum and nuclear domains.

The derivation:

$$h = [PM \times Px \times (2\pi) \times (k-1)^2] / [C \times 7] = 6.6261986282 \times 10^{-34}$$

Measured value: $h = 6.6260755 \times 10^{-34}$

Physical interpretation: Planck's constant is "the ratio of the breaking down of the spherical coherent state PM to the linear displacement rate C."

What this means: - h is not an arbitrary measured constant — it is axiomatically derivable from the PHO framework - It connects the nuclear coherent mass (PM) to the fundamental oscillator rate (C) - The factors (2π) , $(k-1)^2$, and 7 represent the geometric and volumetric conversion from spherical coherence to linear displacement - The match with the measured value is to 7+ significant figures

INW teaching point: "Planck's constant — the most fundamental number in quantum mechanics — isn't just measured. It can be derived from the same axioms that give us proton mass, electron mass, and the speed of light. That's what unification means."

(Source: PHO STATE; PHO Oscillator; PHO-GENETICS — all three derive the same value)

4.10 When Coherence Breaks: Waves, Impedance, and the Photon

When the coherent state of the substratum breaks down, the internal stresses can no longer remain hidden. They transmigrate as waves through the substantial field. The impedance derivation connects the PHO framework to electromagnetic phenomena.

Wave states: - The simultaneous phase transmigrates as C^x - The sequential phase transmigrates as $C^{(1-x)}$ - Their product = $C^x \times C^{(1-x)} = C$ — the total always equals the fundamental oscillator rate

Balanced impedance: - $C^x / C^{(1-x)} = C^{(2x-1)}$ - Numerically: $100 / [(k-1) \times RS] = 377.04$ - This equals the measured impedance of free space in physics (377.0376Ω)

The photon: - When acceleration exceeds C, stress quanta radiate in holographic form - A total of 4.96×10^{17} My counts (= seven Ne's) are needed to initiate the accelerative process as a simultaneous package — this is the photon - The wave is NOT a photon

"radiating in empty space with a velocity" — it is a transmigrating stress in substantial space

INW teaching point: "Light isn't a little ball flying through nothing. It's a stress pattern passing through a real, substantial field — like a ripple moving through water. The water doesn't go anywhere; the pattern does."

(Source: PHOSTATE; PHOscillator; PHO-GENETICS)

4.11 "Dark" Matter and Energy — Sankhya's Precise Explanation

Modern cosmology's "dark matter" and "dark energy" are not mysterious unknown substances. In Sankhya, they are precisely defined as the **unobservable interactive states that are approximately 10^{18} orders of Mly (Moolaprakriti) below h** — the first detectable quantum. The term "dark" simply means: too far below the threshold of h to produce a detectable quantum of change.

The key equation: $Mly \times C^2 = h$

G. Srinivasan establishes that the fundamental mass unit Mly (Moolaprakriti) has an energy value exactly equal to Planck's constant:

$$Mly \times C^2 = h = 6.6261986 \times 10^{-34} \text{ J}\cdot\text{s}$$

(Source: PHO-GENETICS.pdf, p. 4: "my is the fundamental mass value in Sankhya and its energy value is equal to Planck's constant")

This means Mly and h are related through C^2 — the two-axis synchronised count. Planck's constant h is therefore not a standalone arbitrary number but the energy equivalent of one Moolaprakriti scaled by C^2 .

The $\sim 10^{18}$ ratio: gap between Mly and h

The spectrum from Mly (the minimum quantum oscillator) up to h (the first observable quantum) spans approximately:

$$h / Mly \approx 4.93 \times 10^{17} \approx 10^{18} \text{ Mly counts}$$

(SoS1, Eq. [0.52]: $Tq = h/Mly = 4.9278 \times 10^{17}$)

All interactive states below this threshold — hadronic and sub-hadronic counts below h — are permanently hidden: they cannot produce a discrete detectable frequency change. This is the precise Sankhya definition of "dark."

Dark Matter vs. Dark Energy in Sankhya:

Term	Sankhya Definition	Physical Character
Dark Matter	Hadronic unobservable states below h — interactive counts in the hadronic domain that are below the h threshold; cannot radiate; acts as hidden mass/density	Coherent simultaneous counts; acts as inertia/mass but produces no detectable quantum
Dark Energy	Energy equivalent of h in Mly counts — the energy resident across the $\sim 10^{18}$ Mly-count spectrum below h ; the hidden potential of the field	$Mly \times C^2 = h$; the latent energy driving the coherent substratum

The EP8 Value — Energy in the Fundamental Field

The energy resident in the fundamental field as coherent unmanifest activity is:

$$EP8 = Kx / [My \times (RS-1)] = 3.334 \times 10^{-52}$$

This is "the state before the field appears in quantum physics" — the pre-quantum coherent energy of the substratum itself.

"There exists no experimental way of measuring the potential or coherent state of dynamic activity." — G. Srinivasan, SoS2, p. 357

The Hidden Two-Thirds: Thama and Raja as the Perpetual Engine

G. Srinivasan states the practical significance of the Sankhya paradigm with precision:

"The question arises as to what are the advantages of making a paradigmatic change on a vital aspect of Physics. The answer is that exposing two thirds of phenomena hidden from human understanding can give a tremendous motivation to humanity to correct its focus and thereby gain the maximum advantage that the substantial continuum of space can offer, especially in the realm of extracting over-unity energy from the limitless dynamic source." — G. Srinivasan, SoS1, p. 186

The "two thirds of phenomena hidden from human understanding" maps directly to the triadic Guna structure of the Substratum. In the Sankhya framework, all activity resolves into three complementary components — Thama (compressive density), Raja (resonant shuttling), and Sathwa (expansive radiation). Of these three, only **Sathwa** manifests as observable phenomenon: mass/charge states and radiation. The remaining two components — **Thama and Raja** — constitute the hidden, perpetual PHO engine that drives all manifestation but produces no detectable quantum.

The arithmetic is exact: total Substratum activity comprises three Guna components, of which two (Thama + Raja) remain hidden and one (Sathwa) becomes observable. This yields the precise ratio — **two-thirds hidden, one-third observable** — that Srinivasan identifies as the scope of phenomena concealed from human understanding.

In the pre-manifestation Atho state, as described in the 2×3 Guna matrix (§3.2.4), the Sathwa component equals zero. Only Thama and Raja are present — compressive density and resonant shuttling in perpetual balance — with no expansive or radiant mode active. The balance condition $(\pi \times x) / e = (7 \times RS) / 10$ maintains this equilibrium. Manifestation occurs precisely when this Thama-Raja balance is disrupted and Sathwa becomes non-zero, transitioning the Substratum from hidden (Thama-Raja only) to observable (Thama-Raja-Sathwa). The resulting polarization produces either mass/charge states (observable matter) or radiation states (observable energy) — both manifestations within the holographic Substratum.

The "limitless dynamic source" Srinivasan references is the eternal Thama-Raja balance in the Atho state: a perpetual resonance that stands ready to power manifestation anytime, anywhere, across the 10^{18} -order simultaneous interaction spectrum. Understanding this hidden two-thirds engine — rather than treating the unobservable domain as a mysterious "dark" substance — opens the pathway to harmonizing with nature's own wheelwork and extracting coherent energy from the Substratum's perpetual activity.

INW teaching: "What scientists call 'dark matter' is not a mystery substance — it's the hadronic count-activity that exists below Planck's constant h , the first detectable quantum. It acts like mass because it has count-density, but it can't be seen because no interaction in it is large enough to produce a detectable quantum. 'Dark energy' is the energy equivalent of h expressed in Moolaprakriti units: $Mly \times C^2 = h$."

(Source: PHO-GENETICS.pdf, p. 4–5; SoS1, Eq. [0.52]; SoS2, p. 357)

4.12 Why PHO Is Not Perpetual Motion

Modern physics rejects perpetual harmonic motion as violating energy conservation, but this objection assumes empty space and externalised matter. In Srinivasan's framework:

If space is a substantial coherent continuum, then perpetual harmonic oscillation is not a machine violating nature — **it IS nature's own balanced condition.**

The field stays coherent by "internal exchange of oscillatory counts" carried out as a fully internal transfer. Nothing enters or leaves; the exchange is closed and self-sustaining.

"The field stays coherent by internal exchange of oscillatory counts carried out as a fully internal transfer." — G. Srinivasan, SoS2, pp. 425–426

(Source: PHOSTATE.pdf; Key Insights Report; SoS2, pp. 425–426)

4.13 Ferromagnetism as PHO State

The derivation of Iron's atomic characteristics from PHO axioms provides a compelling proof of scale invariance and the practical applicability of the PHO framework:

Magnetic state: Created at the resonant oscillatory rate of 2 cycles per period at maximum coherence.

Property	Axiomatically Derived	Physics Measured	Match
Atomic mass number	55.867	Fe = 55.845	Very close
Element number	26.07	Fe = 26	Very close

"Ferromagnetism is a PHO state at 55 protons & 26 electrons." — G. Srinivasan, PHOSTATE

Why this matters for INW: The same axiomatic method that derives subatomic particle masses can also derive the properties of a specific chemical element — demonstrating scale invariance from nuclear to atomic domains.

(Source: PHOSTATE; PHOscillator; PHO-GENETICS)

4.14 Quick Reference: Three PHO Papers Compared

Users of the Guide may need to consult individual PHO papers. This quick-reference comparison helps orient them:

Document	Primary Focus	Key Contribution	Best Used For
PHOSTATE	Energy and technology claim	PHO as axiomatic state of space; overunity theoretical foundation	Understanding the energy implications; technology applications
PHOscillator	Full numerical architecture	Complete derivation set with Mathcad notation	Detailed numerical verification; checking specific derivations

Document	Primary Focus	Key Contribution	Best Used For
PHO-GENETICS	Scale invariance claim	Same PHO logic applies to hadronic, leptonic, and cell structure	Demonstrating universality; biological extensions

Note: All three papers derive identical numerical values for hadronic masses, leptonic masses, Planck's constant, and the key ratios (PR, RS). They differ in emphasis and framing, not in content.

(Source: PHO Analysis Report)

PART 5: THE TRI-GUNA – THREE MODES OF INTERACTION

5.1 Authentic Definitions

The Tri-Guna are **NOT** moral qualities ("good/bad"). They are the three technical modes of stress interaction that cause all phenomena in the Universe:

Guna	Sanskrit Meaning (G. Srinivasan)	Physical Mode	Mathematical Expression
Thaama	Compressed, contracted, static mass	Compressive, inward, density-forming	$x^2 = 1/2 - x^3/2$
Raja	Shuttling action, interaction, oscillation	Resonant, shuttling, bonding exchange	$x^3 = x - x^2$
Sathwa	Expansion, vector, tensor, radiant state	Expansive, outward, radiant	$x = 1/2 + x^3/2$

(Source: SoS2, Lexicon pp. 494-502; Key Insights Report)

5.2 The Tri-Guna as PHO Phases

Every cycle of interaction expresses all three Gunas:

1. **Thaama** → Squeeze (compressive collision)
2. **Raja** → Exchange (resonant reversal and coupling)
3. **Sathwa** → Release (expansive separation and radiation)

INW phrase: "Every event squeezes, exchanges, and releases."

5.3 Teaching Note

The Gunas are NOT separate forces. They are three inseparable aspects of every single interaction cycle. Like a single breath has inhalation, pause, and exhalation — you cannot have one without the others.

PART 6: KEY SANKHYA CONCEPTS

6.1 Purusha ($K_x \approx 0.9149879$)

Authentic definition: The coherent, static-seeming, central potential state of maximum density. It is the largest coherent unit of cyclic time-period, equated to maximum inertia, mass, or delay as the ground state. In Sankhya it is called the "Andhathamishra" or black-hole state.

Purusha is NOT: - ❌ A "person" or "soul" in the common philosophical sense - ❌ "Consciousness" in the transcendent sense - ❌ A moolaprakriti

Purusha IS: - ✓ A state of maximum density ($K_x \approx 0.9149879$) - ✓ The coherent center around which activity manifests - ✓ The synchronised, static-seeming high-potential state

(Source: SoS2, Symbols & Definitions, p. 504; SoS1, Eq. [1.15])

6.2 Moolaprakriti ($M_{ly} \approx 1.3446 \times 10^{-51}$)

Authentic definition: The smallest cyclic unit of activity — the elemental perpetual quantum oscillator, equated to the fundamental unit of charge. All counts are in terms of the M_{ly} value and therefore dimensionless.

$$M_{ly} = K_x / C^6$$

This is approximately equal to Planck's constant h/C^2 and is consistent with values in Physics.

(Source: SoS2, Symbols & Definitions, p. 504; SANKHYAABSTRACT, p. 9)

6.3 Prakriti and Vikrithi

Term	Authentic Meaning	Physical State
Prakriti	Resonance, oscillatory, dynamic state	Hadronic (nuclear) domain
Vikrithi		Leptonic (electron) domain

Term	Authentic Meaning	Physical State
	Harmonic-oscillator, expansive, change	
Mahad Prakriti (Mps)	Largest dynamic mass = Planck mass	Break in coherent state → start of synchronised state
Mahad Vikrithi (Me)	Reactive/observable unit = electron mass	Reactive phase output when triggered

(Source: SoS2, Symbols & Definitions, pp. 504-505)

6.4 Linga and Bhava

Term	Authentic Meaning	Physical Role
Linga	Coherent state, superposition, field-potential, inertia	The static-seeming coherent field form
Bhava	Kinetic charge, waveform, manifest forms	The dynamic, kinetic, charge-bearing interface

The Linga-Bhava polarisation represents the interface between coherent potential and dynamic manifestation.

(Source: SoS2, Lexicon; SoS2, Fig: The Linga-Bhaava Polarising Form)

6.5 Abhiman and Ahankar

Term	Authentic Meaning	Physical Role
Abhiman	Self-potential, self-measure, self-dimension	The potential well at the resonant interface
Ahankar	Dynamic self-potential, self-action, self-motivation	The spectrum of 7 radiative Ne states; the active self-organising principle

(Source: SoS2, Lexicon pp. 494, 502; SoS2, Fig: The Ahankar Spectrum of 7 Ne States)

6.6 Vrithi (Vortex)

Authentic meaning: Vortex, photon, oscillatory cycle, circular movement.

In the Sathwic (expansive) mode, Vrithis consist of Moolaprakriti (My) units accelerated in groups, packages, or quanta. A total of 4.96×10^{17} My counts (= seven Ne's) are needed to initiate the accelerative process as a simultaneous package — this is the **photon** in Physics.

(Source: SoS2, Symbols & Definitions, pp. 505–506)

6.7 Duhkha = Interactive Stress (NOT "Suffering")

The first decisive correction in Srinivasan's transliteration:

Common Translation	Authentic Meaning
Pain, suffering	Interactive stress; constrained imbalance

This unlocks Sutra 1 as physics of manifestation, not philosophy of suffering.

(Source: SoS2, Lexicon: "dhu:kha = stress, pressure, force, pain")

6.8 The 68 Sutras — Organized by G. Srinivasan

The Sankhya Karika contains 68 axiomatic theorems (Sutras), transliterated by Ishwara Krishna from Maharishi Kapila's science. The following grouping follows G. Srinivasan's authoritative structure in Secret of Sankhya Part 2, pp. 125–140 ("Summary of Suthras").

Organizational note: G. Srinivasan groups the sutras thematically to reveal the logical flow of Kapila's axiomatic chain. Each group builds on the previous one, progressing from the fundamental definition of the substratum through to the complete model of manifestation and liberation.

Why 68 matters: Through the method of elliptical negation in Sutra 1, factorial 68 numbers of solutions can be derived combinatorially. (Source: SoS1, Contents)

Group 1: Fundamental Space (Sutras 1-3)

Theme: Defining the substratum and its foundational properties

Sutra	Sub-heading (G. Srinivasan)	Key Meaning
1	The Dynamic Substratum	Investigating the triad of interactive stresses confirms their existence; such modes would not be detectable without the coherent, perpetual, dynamic, unmanifest state of the substratum (Aikaantha-Athyantha-Atho-Abhavath).
2	Qualities of Substratum	Detection methods have limits (distortion, attenuation, resolution); the alternate method is discriminating the substratum into: unmanifest, manifest, self-potential, and kinetic potential.
3		The fundamental oscillatory state is synchronised, coherent, resonant, stable; the first interactive state = maximum acceleration intensity; seven levels of coherent harmonic

Sutra	Sub-heading (G. Srinivasan)	Key Meaning
	Axiomatic 3D Wave Fundamentals	oscillatory stages; above sixteenth order = expanding radiation; nuclear core = neither oscillatory nor harmonically interactive.

INW teaching takeaway: The first three sutras establish what the universe is made of — a real, dynamic, coherent field that hides itself through perfect balance.

Group 2: Intellectual Verification (Sutras 4-6)

Theme: Methods for proving and knowing reality

Sutra	Sub-heading	Key Meaning
4	Verification of Reality Through Axiomatic Proof	Siddhi (conclusive holistic proof) is arrived at by logical/ theoretical analysis of observations, inferences, and axiomatic principles; condensed further using threefold analytical process = conclusive axiomatic theorem.
5	Process of Verification of Detectable Phenomenon	For persistent sensory perception, there are three information aspects: (positive) detectable, (negative) undetectable, (neutral) original characteristics → measure, analyze, interpret to establish axiomatic theorem.
6	Process of Holistic Derivation of Proof Is Siddhi	For imperceptible, mobile, expansive, undetectable phenomenon → use inferential, holistic, deductive, logical verification technique to mentally experience phenomenon = Siddhi (perfect resonance).

INW teaching takeaway: These sutras establish that there are reliable methods to know what cannot be directly seen — through axiomatic reasoning and the Siddhi process of perfect mental resonance.

Group 3: Axiomatic Reasons for Non-Detection (Sutras 7-10)

Theme: Why the substratum and certain phenomena remain hidden

Sutra	Sub-heading	Key Meaning
7	Reasons Why Manifestation May Not Be Detected	Extremely far/near distances, mental/sensory inefficiencies, subtle/attenuated conditions, occultation/eclipsing, poor background contrast, camouflaging effect.
8		

Sutra	Sub-heading	Key Meaning
	Reason Why the Fundamental Substratum Is Not Detectable	Non-detectability is due to extremely attenuated reactions , not non-existence; only reactions are detectable; when primary displacement occurs → sequence of oscillations detected (original form or harmonics).
9	Logical Reasons Why Substratum Must Exist If Manifestation Exists	Axiomatically illogical to accept continuous/all possible modes of action without a physical cause; the ability to act in all possible ways requires a cause → axiomatic rule.
10	Detection of Manifestation Is Due to Changes	Caused detectable effects of phenomena are: relatively temporary, confined to limited region, proceed action to action (sequential), become cause of further action, have identified characteristics, substantial/with mass, resultant/secondary trait; unmanifest cannot be detected due to inability to discern such effects.

INW teaching takeaway: Something can be real and undetectable at the same time. Non-detection does not mean non-existence — it means perfect balance.

Group 4: Self-Similar Law of Interaction (Sutras 11-14)

Theme: The three Gunas (interactive stress modes) and their self-similar nature

Sutra	Sub-heading	Key Meaning
11	Ability to Discriminate 3 Aspects of Gunas = Cause of Detectability	Inability to discriminate between the triad of stressed states (dynamic connection) = cause of not detecting in synchronised/static/unmanifest state; the three-mode interactive principle applies to first/primary intense state, succeeding reactive states, and nucleus (in opposite/non-detectable state).
12	Characteristics of Gunas (Triad of Interactive Stresses)	Just as human stress transfers through buoyant → calm → despair; Guna interactions = free mobile expansion → balanced resonant interface → compact static contraction; three states mutually interact to override/strengthen/weaken one another, be creative/destructive, associate/combine, exist as self-supporting resonant/dynamic entity.
13	Definition of Gunas	Sathwa = outward operation at maximum rate/intensity/energy transmitting past nuclear boundary beyond first octet (fluidic/flexible state); Raja = transition region of seven oscillatory + first (octet); shuttles inward/outward transferring interactions from nuclear boundary to second radial boundary and vice versa;

Sutra	Sub-heading	Key Meaning
		Thaama = decelerating interaction acting inward forming static nuclear boundary; Vrithi = resultant interaction radiated as self-sustained vortex of radiant energy created by permutations/combinations of previous three levels to transfer interaction/radiate energy.
14	Holistic Logic Can Predict Hidden States	Through holistic analytical derivation of proof: non-detection/detection is due to observer's inability/ability to detect any/all three modes of exchange/transfer of interactions; the Siddhi process itself is due to unmanifest substratum reacting through Guna action by its own inner motivation/potential/cause to manifest as awareness/consciousness.

INW teaching takeaway: Nature's interactions follow three modes — compression, resonance, and expansion. These modes create everything from hidden mass to radiant light. The same three-mode rule works at every level (self-similar).

Group 5: Phases of Oscillatory States (Sutras 15-17)

Theme: How cyclic interaction creates waveforms and phase transitions

Sutra	Sub-heading	Key Meaning
15	Cyclic Oscillatory Interaction Creates Natural Waveform	Cyclic interaction at interface is due to sequentially changing value (acceleration/deceleration); cyclic action = cause and effect by turns to attenuate and concentrate (expand/contract, decrease/increase density) producing oscillatory waveform = standard in nature.
16	First Change of Phase Occurs with Increasing Interaction	In manifestation of decelerating potential → triad of interactions initiated into action → rising/increasing effort/interaction/pressure; a transformation/distinctive change of state occurs (like vapour condensing to liquid) when inward-going interaction brought to very restful (synchronised = static) state.
17	Second Change of Phase Creates Nucleus	Nucleus = location/center/core holding power/potential/interaction/energy from cyclic collision/aggregation due to close contact (back-to-back) of oscillating volumes/waveforms, initiated/triggered by triad of Guna interactive principle in relatively isolated, free, synchronised state. [Summarising: Purusha/nucleus = superpositioning of 3D oscillatory waveforms due to Guna action in static state of dormant potential in undetectable, absolute substratum that is synchronised, coherent, perpetually dynamic but

Sutra	Sub-heading	Key Meaning
		unmanifest; period determined solely by time constant in Sutra 3 forming self-potential in Sutra 2.]

INW teaching takeaway: Oscillations create waveforms. When waveforms collide and synchronize, they form the nucleus (Purusha) — a hidden, dense, high-potential core.

Group 6: Nuclear State as Superposed Coherent Activity (Sutras 18-21)

Theme: The nucleus (Purusha) as a many-component coherent state

Sutra	Sub-heading	Key Meaning
18	Logical Proof That the Nucleus Is a Conglomerate	Since causative action leading to aggregation/dissolution (creation/destruction) are not simultaneous/instantaneous at nuclear interface → holistic conclusion: there must be many nuclei/individual core components (Purusha); also because reversal of triad interactions produce multiple types of phenomenon (should have been singular had nucleus been singular/elementary object).
19	The Complex Nuclear State Forms Background for Manifestation	Conversely inferential holistic conclusion: nuclear state forms background with particulate/inertial mass, relatively seen as neutral, unhindered, static.
20	Static and Dynamic State Is Sustained by Interaction	Because of proximity of static/dynamic states → static state seems dynamic; through Guna action → dynamic seems to behave in static mode maintaining balance.
21	Nuclear State Is Reference Background	Fundamental measurement of phenomenon references nucleus; static nuclear state provides basic background to detect/measure first (primary/fundamental) active (manifested) state as comparative/relative difference; manifestation proceeds on principle of fulfilling need to maintain balance (like blind man and lame man team up to behave normally/effectively).

INW teaching takeaway: The nucleus isn't a single thing — it's many synchronized components acting as one. This complex hidden core becomes the reference background for all measurement.

Group 7: Logic of Self-Sustaining Activity (Sutras 22-25)

Theme: The mathematical derivation of self-similar, perpetual activity

Sutra	Sub-heading	Key Meaning
22	Mathematical Derivation of the Dynamic Self-Similar State	By action of primary interaction → self-sustaining oscillatory state established; from this → series of measurable/detectable energy radiated at sixteenth power of primary value in progressive series incremented to sixteenth level binding/condensing phenomenon through five levels into five sets of manifested phenomenon.
23	Ascertainment by Axiomatic Mathematical Logic	Intellectual logical confirmation process involves application of axiomatic law in expansive mode, backed with knowledge, flexibility to change scales/ratios in order of powers; opposite (contractive, inverted, fractionising concepts) lead to misconceptions/errors of judgement.
24	Mathematical Description of Field Force as a Spectrum	Static potential and kinetic self-similar force (energy) initiate action in progressive incremental value in sequential sets of two → spectrum containing sequential group of eleven, in groups of fives, of field (subtle) forces.
25	Expansive Interaction Rises to 11th Power	Expansive Satwik state has set of eleven levels of change; compressive Thaamasic change initiated through self-action with elemental particles with mass and self-sustaining vortex of charge; interactive Rajasic state is both expansive and compressive.

INW teaching takeaway: The field isn't random — it follows precise mathematical patterns of self-similar growth. These patterns create the spectrum of forces we observe.

Group 8: Cerebral System as a Universal Function (Sutras 26-28)

Theme: Mind, senses, and action governed by the same universal Guna law

Sutra	Sub-heading	Key Meaning
26	Efferent and Afferent Sensory Systems	Efferent (input sensory responses) = light, sound, smell, chemical (taste), contact (touch); afferent (output/action responses) = oscillatory, exchanging, transporting, radiating, creating.
27	Interactions in the Sensory Systems	Mind/cerebral system capable of processing both efferent (incoming input sensory information) and afferent (outgoing action-oriented outputs); can produce specific output despite diversity created by permutations/combinations of internal aspects of three Gunas with numerous external incoming signals.
28	The Five Levels of Cyclic Vortex	

Sutra	Sub-heading	Key Meaning
	(Photonic) Activities	Five types of sensory input signals activated by discrete quantum of forces → results in five categories of output: oscillatory, interactive, transmigrating, radiating, creative activities.

INW teaching takeaway: The mind isn't separate from physics — it operates by the same Guna laws. Sensory input and action output follow the three-mode interaction principle.

Group 9: Perpetual Oscillation and Radiation (Sutras 29-31)

Theme: How cyclic vortices (photons) are formed and sustained

Sutra	Sub-heading	Key Meaning
29	Definition of Self-Similar Internal Activity and Radiation	Cyclic vortex functions on self-similar principles up to third power, is non-synchronous, yet interacts internally together extraordinarily; in synchronous accelerated state at fifth power → becomes fundamental unit of energy radiation.
30	Definition of a Cyclic Vortex and a Spherical Harmonic Oscillator	When oscillatory cycle count in all directions act simultaneously/ synchronously and raised to 4th power → cyclic self-supporting Vrithi (photon) is formed; when it acts non-synchronously/ sequentially → becomes detectable/measurable with degree of certainty; in initial state prior to above when oscillatory count reaches power of three and synchronised/acts simultaneously in all directions → self-supporting harmonic oscillatory state but not detectable/observable.
31	The Cyclic Vortex Is Kept in Continuous Oscillation Only by Its Internal Potential	Cyclic vortex/spherical oscillator kept in continuous interactive exchange up to the very end (limit) only by nuclear/core potential developed by mutual exchange of internally motivated/triggered self-similar/self-organized impulse/force; no other external potential cause.

INW teaching takeaway: Light (photons) and stable matter are both oscillators — they differ only in their level of synchronisation. Matter is 3rd-power synchronised; photons are 4th-power vortices. Both are self-sustaining with no external cause.

Group 10: Potential as Coherent States (Sutras 32-34)

Theme: How potential builds up through coherent synchronisation

Sutra	Sub-heading	Key Meaning
32	The Static and Kinetic Potential Limits	Potential rises to 13th power to accelerate, superpose, radiate; consequently kinetic potential rises to 10th power to accelerate, superpose, radiate.
33	The Internal and External Potential Limits	Internal limit of potential is at third power; externalising detectable factor (existing in present time) is up to 10th power; third-power factor contributes to internal bonding or is a limiting state.
34	The Potential to Act Is at the 5th Power Level	Will/internal potential to act increases rate to 5th power of both specific/non-specific (observable/hidden) sensory signals; sound signals relate to speech/oscillatory functions; rest relate to all five observable interactions.

INW teaching takeaway: Potential isn't vague — it has precise power-law levels. The 3rd power creates bonding/internal coherence. The 5th power creates the will to act. The 10th-13th powers create radiation.

Group 11: Third-Order Constraint Causes Phenomena (Sutras 35-39)

Theme: The third-power damping constraint and its consequences

Sutra	Sub-heading	Key Meaning
35	The Formation of a Basic Field Is Determined by the 3rd Order Damping Action	At point when all measurable interactions plunge to limit and form coherent bond = measure of effort involved; therefore third power of this bond forms base/limiting medium; remaining powers of interaction use this as medium.
36	Decay of Radiated Phenomenon	Spectrum of light transmission process = complex/mutually interactive Guna exchange sequence; totally controlled by Purusha potential; it is the only coherent motivating factor clearly till its end.
37	Extraction of Potential Energy from the Purusha Domain	Comprehensive/extraordinary use can be made in following way: Purusha potential can be attained from minute coherent state hidden inside Purusha by repeated precisely triggered inputs to initiate primary interactions.
38	Classification of Activity of Vrithi and Elements	Self-generated vortex (Vrithi) has non-specific/wide range of activity level; elements are at five levels up to fifth power of activity; spectrum logically covers classes of activity states: synchronised, non-synchronised, interactive, superposed, coherent.
39	Genetic Code	

Sutra	Sub-heading	Key Meaning
		Field sustained by fundamental source of energy bonds into three organic species; mobile field is constant and controlled; organic states of three species are cyclic/periodic.

INW teaching takeaway: The third power is the damping constraint that prevents collapse or runaway expansion. This creates the base field. From this base, higher powers create radiation, vortices, and even organic life patterns (genetic code).

Group 12: Superposed States Create Ground Potential (Sutras 40-42)

Theme: How static mass states are formed through superposition

Sutra	Sub-heading	Key Meaning
40	Linga State as Coherent and Superposed Oscillations	Static mass states created by maximally absorbing/superpositioning entire spectrum of vibratory/oscillatory states, from largest/strongest to (limit) finest minimal kinetic charged states.
41	The Need for the Substratum as Supporting Framework for Manifestation	Just as it is not possible to present picture without supporting base nor cast shadow without appropriate pillar → impossible to have synchronised/superpositioned static state without required supportive base.
42	The Synchronised State Provides Attracting Potential to Build Up Mass	Synchronised/coherent state of nuclear core forms target potential to attract oscillatory interactions to synchronise/superposition itself circumferentially, simultaneously, similar to movements in a dance, which increases/strengthens resonant state of interactions.

INW teaching takeaway: Mass isn't a property particles "have" — it's what happens when oscillations synchronize and stack up (superposition). The coherent core attracts more oscillations into synchrony, building up mass.

Group 13: The Balance in Interactive States (Sutras 43-45)

Theme: How balance determines detectable vs. undetectable states

Sutra	Sub-heading	Key Meaning
43	Conditions Under Which the Laws of	Source/starting point of axiomatic laws of resonant action are from state when charged interactive state is in perfect resonance; starting point/start of time-cycle-period count of

Sutra	Sub-heading	Key Meaning
	Interaction Are Certain	measurement commences when action is in state of relative rest/static state (detection only possible when detected component is at restful state).
44	The Mode of Natural Action Based on Rules	Satwic expansive actions supported by axiomatic laws and expand towards free state; actions opposing it result in reduced states leading to Thaamasic/compressive states that are restrictive; inner self-potential is only cause of synchronisation/establishment of coherent state; obstruction/restrictions result in confined/bonded state.
45	State of Unbalance Is the Motivating Cause of Manifestation	When Rajasic interactive states are equal and balanced → accumulation/increase of mass by absorption/superpositioning of active displacements occur; when unbalanced/unequal → increase/decrease in force causes acceleration/deceleration creating manifest spectrum of universal phenomenon.

INW teaching takeaway: Perfect balance creates mass (accumulation). Imbalance creates motion and radiation (manifestation). The laws are certain and deterministic when we measure from a restful reference state.

Group 14: Spectrum of Interactions Defined (Sutras 46-50)

Theme: The complete power-spectrum of manifestation (up to 50 orders)

Sutra	Sub-heading	Key Meaning
46	Interactions in the Synchronised State Cause 50 Orders of Change	Ascertained order of spectrum of interactive states due to conditions of unbalance, non-synchronisation, balance, coherent synchronisation caused by interplay of three Guna modes of interaction rise up to 50 (power index).
47	Balanced, Synchronised and Its Reversed Variation Sequence of Interactions	Sequence of 5 orders (power index) of change created by three varying states: non-synchronous, interactive change, synchronous; weak/unsynchronised states have 28 orders; balanced phase has 9 orders; perfectly synchronised state has 8 orders of sequentially variable levels.
48	Order of Compression Increase When One, Two, and Three Axes Synchronise	Interactive changes in compressive Thaamasic state proceed on basis of 8 orders of change along one axis; increases by same order when synchronises with another axis at deeper level; at next intermediate level increases by 10 orders; at final core level increases by 18 and another 18 at deepest core level synchronisation → 8 + 5+4 followed by (5) in each axis and finally (8+10) 18; combinatorial process of incremental rates

Sutra	Sub-heading	Key Meaning
		can synchronise/remain in balance only as even-numbered levels.
49	The 11 & 17 Orders of the Weak and Balanced Groups	Eleven orders of expansive interactions cover observable range out of previous 28 orders mentioned before; balance of 17 orders consists of balanced/coherent states and radiant states; with 5 orders of radiant state forms 22 orders; 17 orders form common potential to both phases.
50	Nuclear Balancing Parameters	Interaction inside nuclear state can be defined as: at fourth power, initiating cause of action and responsive self-reaction, time period of reaction, destined potential available for reaction; externally detectable fundamental activity categorised as being at fifth power; when these two act together synchronously = ninth power.

INW teaching takeaway: The entire observable and unobservable universe is a spectrum of 50 power-levels of interaction. The weak/expansive domain (28 orders), the balanced domain (9 orders), and the coherent domain (8 orders) — all follow precise combinatorial rules.

Group 15: Interactions Cause Stress (Sutras 51-56)

Theme: How colliding interactions create vibratory stress; the dual polarisation into mass and charge

Sutra	Sub-heading	Key Meaning
51	Colliding Interaction Cause of Vibrations	Knowledge gained through research on vibratory/oscillatory stress caused by colliding interactions follow three-step action (compression-shuttling-expansion Guna mode) leading to intensive superpositioned, divergent, or synchronised states raised to eighth power in coherent mode; original state prior to interaction established to be in controlled, compressed, cubic, volumetric state raised to third power.
52	The Polarisation of Phenomenon into Coherent Potential (Mass) and Kinetic Potential (Charge) Modes	Neither can a characteristic potential source exist without a kinetic phase nor can a potential phase without a kinetic mode; vorticular particle is initiated by a potential source; potential is classified as mass and kinetic phase as charge; from this dual combination all phenomenon is initiated/maintained by dual mode of polarisation.
53	Guna Classification	Natural/inorganic/matter-oriented phenomenon = result of 8 orders of variations; organic/sub-human order is fivefold;

Sutra	Sub-heading	Key Meaning
		human order is from single variant; these three classes succinctly control complete spectrum of phenomenon.
54	The Distribution of Guna Interactive States	Ascending order of outward manifestation predominantly Satwic; descending inward mode of creation predominantly Thaamasic; intermediate range predominantly Rajasic; consistently so right from starting dynamic creative field to final coherent static order of manifestation.
55	Stress in Three Modes Follows Self-Similar Laws in the Dynamic Substratum	Therefore process of decay/absorption introduces interactive stresses that maintain dynamic state of Purusha; on its absorption of vorticular interactive activity by nuclear core → process of superpositioning continues until interactions minimised and coherent self-similar state established.
56	Manifestation Is the Result of Interactions to Maintain the Balance Between Nucleus and Boundary	In this way interactive oscillatory activity initiated/perpetuated from intense fundamental activity at source to final coherent superposed massive state at isolated nuclear boundary where self-potential balances with reactive potential at starting point.

INW teaching takeaway: All phenomenon comes from dual polarisation: mass (coherent potential) and charge (kinetic potential). The three Guna modes create the stress patterns that build up, transmigrate, and release energy in precise self-similar ways.

Group 16: Coherence Causes Potential States (Sutras 57-60)

Theme: How the nucleus forms through stress absorption and internal motivation

Sutra	Sub-heading	Key Meaning
57	Attainment of a Coherent Superposed State Initiates Oscillatory State	Growth of calf is due to flow of milk that seems to have commenced without cause/specific instruction; similarly formation of nucleus takes place by stress transmigration of static potential through gradual attenuation of primary kinetic force into synchronised state of freedom by its own internal motivation/initiating activity to balance forces.
58	The Unmanifest State Maintains the Potential to Act	People are motivated into maintaining state of activity to reduce their zeal for action; similarly unmanifest state maintains Purusha in isolated state free of stresses by absorbing stresses of continuous static-through-kinetic balancing interactions.

Sutra	Sub-heading	Key Meaning
59	Forming of Nucleus Due to Reduction of Potential Difference	Just as external exhibition of dancer's performance reduces his urge to continue his performance because it satisfies his desires; similarly outward spreading of internally motivated self-energy diminishes its potential to radiate and thereby reaches interactive state of balance in oscillatory state.
60	Cause of Interactive Oscillations	Various factors (controllable, corrective, supportive, non-supportive nuclear state) along with interactive Guna qualities and its opposite coherent superposed states (dynamic and non-dynamic, potential and lack of potential) maintains oscillatory activity.

INW teaching takeaway: The nucleus forms by its own internal motivation — not from an external cause. Like milk flowing to feed a calf or a dancer's energy finding expression, the field's own potential seeks balance. This self-organization is the engine of manifestation.

Group 17: Synchronisation Causes Holographic States (Sutras 61-62)

Theme: The holographic nature of manifestation; nothing truly moves, only patterns transmigrate

Sutra	Sub-heading	Key Meaning
61	The Oscillatory Displacement at the Point of Balance Is Very Attenuated	At nuclear boundary oscillatory displacements reduce to such minute angular oscillatory movement that its very existence seems doubtful; consequent reduction in interactive self-potential does not ever allow it to radiate detectable information on state of interaction of nuclear region.
62	Interactions Are Cause of Manifestation as a Holographic Phenomenon	Therefore nothing is bound, released, or translocated; only the oscillatory waveforms of interactions is bound/ superposed by synchronisation, released by desynchronisation, and transferred by transmigration due to unequal displacements; this dynamic state continues endlessly.

INW teaching takeaway: Nothing actually moves from place to place. What "moves" is the pattern of oscillations. Manifestation is holographic — it's a transmission of synchronized waveform patterns, not a transport of objects.

Group 18: Self-Similar Interactions (Sutras 63-64)

Theme: The seventh-power compression/expansion dynamic and the ground state

Sutra	Sub-heading	Key Meaning
63	Interaction	Oscillatory waveforms superposition itself by its own potential to form nuclear density by compressing its waveforms to seventh power when it becomes equal to static nuclear potential; when it expands it releases one waveform at a time.
64	The Lack of Specific Self-Potential Leads to a Ground State	Hence fundamental research indicates that non-availability of self-potential is cause of existence of potential sink (ground state); not due to any precise measure of interactive control that a pure changeless ground state exists (in nuclear core of components of substratum).

INW teaching takeaway: Compression goes to the seventh power — then it matches the nuclear potential and stops. Expansion releases one waveform at a time. The ground state isn't a "thing" — it's the condition when self-potential is absent (potential sink).

Group 19: Equalisation of Potentials (Sutras 65-66)

Theme: The state of balance when the nuclear boundary is in perfect equilibrium

Sutra	Sub-heading	Key Meaning
65	The State of Balance When the Nuclear Boundary Is in Balance	When oscillatory interactions minimised because of reduction in activity between boundary and nuclear core → interactive current becomes neutral and activities at seven radial levels become insignificant → Purusha attains firm background state.
66	State of Balance by Eliminating the Potential to Act	Observing neutral state of one gives proof that observable movement of the other has ceased; even though both are together no interactive manifestation exists.

INW teaching takeaway: When the nuclear boundary fully balances with the core, all radial activity becomes insignificant. The Purusha becomes a firm, undetectable background. This is the state of liberation — no manifestation, only existence.

Group 20: Conversion of Charge to Mass (Sutras 67-68)

Theme: How kinetic force (charge) converts to static potential (mass) and vice versa

Sutra	Sub-heading	Key Meaning
67	Conversion of Kinetic Force into Static Potential as Inertia or Mass	Basic principle of acquiring solid state/mass is by synchronising components into relatively static level by allowing spin angular momentum to be converted to potential state; exchange/accumulation is through self-similar proportion or ratio.
68	The Final Synchronised, Coherent, Dynamic State of Isolated Nucleus	Process of acquiring mass is given effect by superpositioning interactive vortex-like waveforms by changing it into gradually diminishing cyclic interaction that reaches synchronised/coherent state of balance within first/primary activity boundary; attaining synchronised/coherent state eliminates delay, inertia, friction and allows this state to continue unhindered in complete freedom.

INW teaching takeaway: Mass is frozen spin. Charge is unfrozen motion. The conversion between them follows the self-similar ratio (golden mean). When synchronisation is perfect, the state is free — no friction, no delay, perpetual.

Structural closure: Sutra 68 repeats the key terms from Sutra 1 — "synchronised, coherent, dynamic" — confirming that the entire 68-sutra treatise is a self-consistent logical loop returning to its starting point. (Source: SoS2, p. 44, Tribute to Kapila)

Group 21: Hierarchy of Knowledge Transmission (Sutras 69-70)

Theme: The lineage of Sankhya knowledge (meta-level sutras)

Sutra	Sub-heading	Key Meaning
69	The Knowledge of the Nuclear Potential Is Codified in This Work	The knowledge of nuclear potential is encoded in this work by the great Maharishi wherein the method of intellectually ascertaining the process of manifestation of phenomenon from its origin, through its growth, and till its dissolution are explained.
70	The Hierarchy of Information Transmission	This pure and exotic doctrine the sage gave to Asuri who in turn handed down to Panchasikha by whom it was extensively propagated as the perfect set of principles explaining the mode of action in reality.

Note on Sutras 69-70: These are meta-level sutras about the knowledge lineage itself (Kapila → Asuri → Panchasikha → Ishwara Krishna). They are not part of the physics/manifestation model. Some scholars count 68 sutras (Groups 1-20) as the main body; others count 70 including the lineage sutras.

INW teaching takeaway: Sankhya is not a religion or mythology — it is a codified science transmitted through a verified lineage of sages who tested and confirmed it through the Siddhi process.

Summary: The Arc of the 68 Sutras

Arc Phase	Sutra Range	Conceptual Flow
Foundation	1-10	Define the substratum; establish methods of proof; explain why it's hidden
Dynamics	11-25	Three Gunas; oscillatory phases; self-similar mathematical derivation
Application	26-39	Mind/senses; perpetual oscillation; radiation; third-order constraint; genetic code
Coherence	40-56	Superposition creates mass; balance vs. imbalance; 50-order spectrum; mass/charge polarisation
Liberation	57-68	Self-organization; holographic manifestation; waveform compression/expansion; balance; conversion of charge to mass; final freedom
Lineage	69-70	Knowledge transmission hierarchy

(Source: SoS2, Summary of Suthras, pp. 125-140)

PART 7: SIDDHI — DIRECT KNOWING THROUGH RESONANCE

7.1 Authentic Definition

Siddhi = resonant, proven, verified, contained skill — the process of perfect resonance with the coherent field, enabling direct experiential knowing.

(Source: SoS2, Lexicon p. 500: "siddhi = resonant, prooven, verified, contained skill")

7.2 The Four Stages

Stage	Sanskrit	Function
Dharana	Collection	Gather the field of attention
Dhyana	Focusing	Bring inquiry into steady coherence
Samadhi	Equality/resonance	Mind becomes equal to the object of inquiry
Samyama	Synchronised result	Solution appears as a complete, internally verified whole

(Source: MEDITATION.pdf; SIDDHI.pdf; Key Insights Report)

7.3 Scientific Basis (Not Mystical)

The mind is a local expression of the same universal substratum. When the body and mind settle into balance, the cerebral field can synchronise with the coherent field of space.

- ✓ This is a lawful process in Sankhya
- ✓ Conditions can be specified
- ✓ Results can be predicted and tested
- ✓ Anyone can develop this capacity
- ✗ Not requiring faith or belief
- ✗ Not dependent on special powers
- ✗ Not supernatural or magical

"What Maharishi Kapila had demonstrated through axiomatic mathematics was that all phenomena in nature was of a holographic or coherent vibratory form. Therefore, the human mind too could reproduce every phenomenon within itself by the simplest process of bringing it to a coherent and synchronised phase." — G. Srinivasan, SoS2, Preface, p. 13

7.4 The 21-Minute Cycle

Practical meditation instruction references a 21-minute cycle and 3.75 Hz Theta/Ishwara-field frequency, connecting physiological rhythm with Earth-field resonance.

(Source: MEDITATION.pdf; SIDDHI.pdf)

PART 8: SANKHYA vs. MODERN PHYSICS

8.1 Key Differences

Aspect	Modern Physics	Sankhya
Space	Empty vacuum	Substantial, dynamic, coherent substratum
Foundation	Empirical measurement	Axiomatic derivation
Method	Hypothesis → Test → Revise	Axiom → Derive → Verify
Certainty	Probabilistic (uncertainty principle)	Axiomatic (counting is certain)
Origin of constants	Measured, not explained	Derived from axioms
Forces	Separate domains (electromagnetic, gravitational, strong, weak)	No separate forces — only different levels of simultaneous interactions
Growth patterns	Observed, catalogued	Derived as mathematically inevitable
Perpetual states	Forbidden (energy conservation)	Natural (PHO state in substantial continuum)
Dark matter/energy	Mysterious unknown substances	Hadronic unobservable states $\sim 10^{18}$ orders of Mly below h; $Mly \times C^2 = h$
Quantum behavior	Paradoxical mysteries	Explained by substratum properties
Electron	Independent particle	Boundary state of hadronic core
Planck's constant	Measured, not derived	Axiomatically derived from PM state

8.2 Sankhya's Explanatory Achievements

Per G. Srinivasan's derivations:

1. **Speed of light (C)** — Derived axiomatically as $C = 10^{(2/x^3)} = 296,575,967$ counts/cycle
2. **Electron mass (Me)** — Derived as Mahad Vikrithi = 9.11023×10^{-31}
3. **Planck mass (Mps)** — Derived as Mahad Prakriti = 2.20369×10^{-8}
4. **Permittivity (ε) and Permeability (μ)** — Derived from impedance relationships
5. **Gravitational phenomenon** — Primary cause: different levels of simultaneous interactions
6. **EPR paradox** — Resolved through simultaneity in substantial substratum

7. **Hubble's redshift** — Not expansion; frequency shifts are logarithmically proportional to distance in a gravitational field
8. **Pioneer anomaly** — Predicted shortfall in stress transmigratory distance
9. **Planck's constant (h)** — Axiomatically derived from PM state: $h = 6.6261986282 \times 10^{-34}$
10. **Neutron mass** — Derived to 7+ significant figures: $1.6749276458 \times 10^{-27}$
11. **Proton mass** — Derived to 7+ significant figures: $1.672621512 \times 10^{-27}$
12. **Nuclear gyromagnetic ratio (Gmr)** — Axiomatically derived as PHO Ratio PR = 3.571428
13. **Iron atomic mass number** — Derived as 55.867 (measured: 55.845)
14. **Free-space impedance** — Derived as 377.04 Ω

(Source: SoS1, Axiomatic Derivations; SANKHYAABSTRACT, pp. 5-8; PHO STATE; PHO oscillator; PHO-GENETICS)

PART 9: TERMINOLOGY STANDARDS FOR INW

9.1 Terms to Use (Authentic to Sankhya)

Term	Usage	Context
Sankhya	The authentic science	"In Sankhya science, we derive..."
Axiom	Foundational provable truth	"The axiom $1+1=2$..."
Derive	Work from first principles	"Derive the golden ratio from..."
Dynamism	Self-sustaining rhythmic interchange	"Nature's just-right dynamism ratio"
PHO state	Perpetual Harmonic Oscillator	"The PHO state is the engine of the substratum"
Substratum	Fundamental substantial space	"The substratum is coherent and dynamic"
Simultaneity & Sequence	Dual modes of activity	"Simultaneous and sequential..."
Tri-Guna	Three modes of interaction	"Compression, resonance, expansion"
Self-similar	Internally generated proportional rule	"The rule comes from within"
Scale-invariant	Same dimensionless law at each level	"The same law applies everywhere"
Coherent state		

Term	Usage	Context
	Many interactions acting as one	"A coherent state forms when many synchronise"
Golden mean conjugate	$x = 0.618034$	"Nature's dynamism ratio"
Siddhi	Direct knowing through resonance	"Siddhi is a lawful process"
Swabhava	Self-nature; internally generated law	"The rule comes from within the interaction itself"
Duhkha	Interactive stress	"Stress drives manifestation"
Counting	The method of Sankhya	"Counting interactions is certain"
Ladder of Phenomenon	C-power states of manifestation	"Each level on the Ladder represents a power of C"
Detection Principle	Only change can be detected	"Non-detection is not non-existence"

9.2 Terms to Avoid or Qualify

Term	Issue	Alternative
Samkhya	Non-authentic spelling	Use "Sankhya" (G. Srinivasan's spelling)
Believe	Implies faith	Use "derive," "verify," "understand"
Spiritual/Mystical	Vague, not scientific	Use "axiomatic," "mathematical," "derived"
Consciousness (capitalised)	Implies transcendence	Use "awareness," "knowing," "coherent state"
Energy (undefined)	Vague physics term	Specify: potential, kinetic, interaction, stress
Vibration (without mechanism)	Vague	Use: oscillation, interaction, resonance
Ancient wisdom	Implies reverence over reason	Use "axiomatic science" or "Sankhya"
Forces (as separate domains)	Implies separate sources	"Different levels of simultaneous interactions"

9.3 Red Flag Language

✗ Avoid	✓ Use Instead
"Scientists believe..."	"From first principles, we derive..."
"This is ancient wisdom, so..."	"Let's verify this by..."
"You just have to understand that..."	"Here's how we derive it..."
"It's spiritual, so..."	"This is a mathematical principle..."
"Eastern philosophy teaches..."	"G. Srinivasan's Sankhya system shows..."
"Energy flows through..."	"Interactive stress transmigrates from..."
"The universe is expanding..."	"Space comprising substantial components cannot expand"

PART 10: TEACHING ANALOGIES FOR INW

10.1 The Clap Analogy (Simultaneity)

Ten claps one after another → countable as ten. Ten claps all at once → countable as one, but with ten times the intensity. Nine counts "vanish" into hidden density. This is how mass, potential, and coherence arise.

(Source: SoS1, pp. 8-9)

10.2 The Archer's Bow (PHO State)

A rigid bow with a string attached: the outer curve stretches while the inner surface compresses simultaneously. The string resonates at a balanced frequency — perpetually. Neither the bow nor the string changes its state. This is the PHO state.

(Source: SoS1, Overview; SoS2, Preface)

10.3 The Train Analogy (Simultaneous vs. Sequential)

Wagon length L = rigid (simultaneous). Gap G = variable (sequential). Both ends of each wagon act simultaneously. The gap between wagons is the sequential interval. Changes in velocity alter the Lt/Gt ratio instantly, requiring a balancing process: this is why $1/x = 1 + x$.

(Source: SoS1, pp. 8-9)

10.4 The Porous Pot Immersed in Water (Siddhi)

Imagine a porous pot immersed in water. If you keep shaking it all the time, very little water seeps into the pot. But if you let the pot remain absolutely still, water from all sides simultaneously seeps in and fills it in the shortest time. We — as bundles of Moolaprakriti — are exactly like that porous pot: immersed in a resonant sea of energy. The more still and coherent we become, the more fully we absorb from the field. The Siddhi process is the deliberate cultivation of that stillness.

"Imagine a porous pot immersed in water. If you keep shaking it all the time very little water would seep into the pot. On the other hand, if you let the pot remain absolutely still, water from all the sides would simultaneously seep into the pot and fill it in the shortest time. Recalling that we in effect are a bundle of Moolaprakritis, just like a porous pot, we must learn to sit very still indeed to absorb from the resonant sea of energy around us." — G. Srinivasan, SoS2, p. 449

(Source: SoS2, p. 449)

10.5 The Moolaprakriti Count Ladder (Coherent Simultaneity)

The "C-power ladder" is most precisely understood as a spectrum of **simultaneous Moolaprakriti counts** — describing how many Mly units are acting coherently at the same instant. The smallest possible count is a single Mly — one elemental quantum oscillator acting alone. As more and more Mly units synchronise and act simultaneously, they form progressively higher-density coherent states. The maximum possible simultaneous count is Kx (Purusha) ≈ 0.9149879 — the upper boundary of coherent activity.

The full spectrum from Mly (one Mly acting alone, the smallest) to Kx (maximum coherent simultaneous count) defines the complete range of manifestation. Every stable physical state — Moolaprakriti, neutrino, electron, proton, neutron, Planck mass — is a specific count of how many Moolaprakritis are acting simultaneously as one. This is what the ladder measures: not floors in a building, but the number of Moolaprakriti acting in coherent unison.

The spectrum: - **Bottom rung:** 1 Mly acting alone = Mly = 1.3446×10^{-51} (minimum quantum unit) - **Upper rung:** Kx worth of Mly units acting simultaneously = Purusha state ≈ 0.9149879 (maximum coherent count) - **Known particles** sit at specific counts between these extremes — each one a characteristic simultaneous Mly count

INW teaching point: "How many Moolaprakritis are clapping at once? One clapping alone is the bottom rung. Maximum coherence — all acting as one — is the Kx top. Every

particle in nature is a specific rung on that ladder: a precise number of Moolaprakritis all beating together."

(Source: SoS1, Axiomatic Concepts; SoS2, pp. 361–362; SANKHYAABSTRACT, p. 9)

10.6 The Drum Surface (Electron as Boundary)

The electron isn't a little ball orbiting the nucleus — it's the vibrating surface of the nucleus itself. Like the surface of a drum isn't separate from the drum body. The same ratio that governs the proton-neutron relationship also governs the electron relationship. One rule, expressed at two levels.

PART 11: COMMON MISCONCEPTIONS & CORRECTIONS

#	Misconception	Correction	Teaching Point
1	"Golden ratio appears because it's beautiful"	ϕ is the ONLY ratio that achieves balanced dynamism. Beauty is a consequence, not a cause.	"Nature uses ϕ because it's the only way to maintain balance while growing."
2	"The 68 axioms are spiritual teachings"	They are mathematical theorems — provable and logical.	"These aren't beliefs. They're mathematical principles you can derive and verify."
3	"Siddhi is supernatural"	Siddhi is a lawful process of resonance with the coherent field.	"Siddhi happens naturally when specific conditions are met."
4	"Sankhya is ancient religion"	Sankhya is authentic science with deeper foundations than modern physics.	"It's rigorous science, not belief."
5	"Space is empty"	Space is a substantial, dynamic, coherent, holographic medium.	"The void is the greatest error in physics."
6	"Forces are separate domains"	There are no separate forces — only different levels of simultaneous interactions.	"What we call gravity, electromagnetism, etc. are all the same thing at different densities."
7	"Perpetual motion is impossible"	In a substantial continuum, PHO is nature's balanced condition.	"The substratum never stops oscillating — it's what 'being' means."
8	"Dark energy is mysterious"	"Dark" = unobservable states $\sim 10^{18}$ Mly counts below h ; Mly $\times$	"It's not dark — it's below the detection threshold of h ."

#	Misconception	Correction	Teaching Point
		$C^2 = h$; Dark Matter = hadronic counts below h; not a mysterious substance.	
9	"The electron is an independent particle"	The electron is the boundary state of the hadronic nuclear core.	"It's the drum surface, not a separate ball."
10	"Planck's constant is just a measured number"	h is axiomatically derivable from the PHO framework.	"Even 'fundamental' constants have reasons."
11	"Nuclear physics is uncertain"	At the atomic/nuclear level, there is no uncertainty — only precise counting.	"Counting is certain. Only measurement is uncertain."

PART 12: QUICK REFERENCE TABLES

Key Values

Concept	Value/Formula	Significance
Axiom	$1 + 1 = 2$	Foundation of everything
Balance equation	$1/x = 1 + x$	Dynamism requires ratio = increment
Golden mean conjugate	$x \approx 0.618034$	Nature's dynamism ratio
Golden mean	$\phi \approx 1.618034$	Expansion ratio
$\sqrt{5}$ connection	$\sqrt{(1^2 + 2^2)}$	Links ϕ back to axiom
Third-order damping	$x^3 = x - x^2$	Prevents collapse or runaway
Ten-count cycle	10 interactions/ cycle	Axiomatic cyclic unit
Purusha (Kx)	0.9149879	Maximum density boundary
C (count rate)	296,575,967	Axiomatic interactive rate
Moolaprakriti (Mly)	1.3446×10^{-51}	Smallest quantum unit
PHO Ratio (PR = Gmr)	$100/28 = 3.571428$	Axiomatically derived; equals the Gyromagnetic Ratio
PHO Resonant Ratio	$100/98 = 1.020408$	Perpetual restoring factor
k (cube root 2)	1.259921	Volumetric increment ratio

Concept	Value/Formula	Significance
Planck's constant (h)	$6.6261986282 \times 10^{-34}$	Axiomatically derived from PM
Free-space impedance (Si)	377.04	Derived from PHO framework

Hadronic Triad

State	Symbol	Derived Value	Physics Measured
Neutron	Pn	$1.6749276458 \times 10^{-27}$	$1.67492728 \times 10^{-27}$
Coherent mass	PM	$1.6744231791 \times 10^{-27}$	Not known in physics
Proton	Pm	$1.672621512 \times 10^{-27}$	$1.67262171 \times 10^{-27}$

Leptonic Triad

State	Symbol	Derived Value	Physics Measured
PM-derived lepton	Mep	$9.1140580241 \times 10^{-31}$	Not predicted
Planck-derived lepton	Me	$9.110233722 \times 10^{-31}$	Not predicted
Balancing state (electron)	Mee	$9.1093838239 \times 10^{-31}$	$9.1093897 \times 10^{-31}$

PART 13: CORE PRINCIPLES — WHEN IN DOUBT

- ✓ **Start with 1+1=2** — Everything traces back to this axiom.
- ✓ **Derive, Don't Believe** — Show the logical steps. Never ask readers to accept without derivation.
- ✓ **Nature Has Reasons** — Every pattern serves a purpose. The golden ratio isn't arbitrary — it's inevitable.
- ✓ **Count, Don't Assume** — Counting interactions is certain. Time and measurement are relative.
- ✓ **PHO Is the Engine** — The substratum never stops oscillating. All forms arise from balanced/broken/restored PHO.
- ✓ **Invisible Is Not Unreal** — Non-detection does not equal non-existence.

✓ **Unify, Don't Separate** — There are no separate forces, no separate domains. Everything is one field at different densities.

✓ **Empower, Don't Lecture** — Help readers discover. Don't tell them what they should find.

✓ **Sankhya Deals Only With Reality** — No speculation, no belief, no mysticism. Only axiomatically derived truth.

✓ **Follow the Ladder** — From axiom to Kx to C to My, every value connects.

✓ **The Same Ratio Everywhere** — PR (= Gmr) and RS appear at hadronic, leptonic, and atomic levels.

Document Status: v2.4 — CANONICAL-SOURCE & REGISTRY ALIGNMENT **Version:** 2.4 (July 16, 2026 — aligned to canonical SoS1/SoS2 + Master Symbol & Terminology Registry v1.0; §2.8a two light speeds; Appendix A extended derived constants; see Changelog) **Last Updated:** July 16, 2026 **Source Authority:** Exclusively G. Srinivasan authentic works (SoS1 189 pp. + SoS2 571 pp., locked in-repo under `sources/`) **Companion documents:** Master Symbol & Terminology Registry v1.0 (symbol/value authority) · Sankhya Glossary v2.3 · Teacher Guide QA Checklist v1.0 **Maintained By:** INW Project Content Team

APPENDIX A: EXTENDED DERIVED CONSTANTS (VETTED)

This appendix lists additional physical constants that Sankhya derives axiomatically from the cyclic variable $\mathbf{x} = \mathbf{0.618034}$ and the count-rate $\mathbf{C} = \mathbf{Cs} = \mathbf{296,575,967}$. Every value below has been cross-checked against the SoS1 equation set and recorded in the **Master Symbol & Terminology Registry v1.0 (§3)**, which holds the authoritative sourcing and equation numbers. Values are quoted as derived; the closely matching modern (CODATA) quantity is given for orientation only.

A.1 Core cyclic and field constants

Symbol	Name	Derived value	Modern equivalent
$\mathbf{C} = \mathbf{Cs}$	Axiomatic count-rate	296,575,967 (2.9657596692×10^8)	— (axiomatic "speed of light")
$\mathbf{c} = \mathbf{CL}$	Measured light speed	299,792,458	Speed of light on Earth ($= \mathbf{Cs} \times \mathbf{Fc}$)

Symbol	Name	Derived value	Modern equivalent
Fc	Blue-shift correction	1.010845	CL / Cs
Gm	Base gyromagnetic count	7.142857 (= 100/14)	—
PR = Gmr	PHO / gyromagnetic ratio	3.5714286 (= Gm/2 = 100/28)	Nuclear gyromagnetic ratio
Rs	Resonant restoring ratio	1.0204081633 (= 100/98 = Gm/7)	—
K7 = Tce	Seventh count / thermal-cyclic exponent	26.9312547	—
Kx	Purusha constant	0.9149879388	≈ Catalan's constant

A.2 Planck-scale constants

Symbol	Name	Derived value	Modern equivalent
h	Planck's constant (= 7·Ne)	$6.6701138380 \times 10^{-34}$	$6.62607015 \times 10^{-34} \text{ J}\cdot\text{s}$
Lp	Planck length	$1.6895595727 \times 10^{-35} \text{ m}$	$1.616255 \times 10^{-35} \text{ m}$
Tp	Planck time	$5.6968863333 \times 10^{-44} \text{ s}$	$5.391247 \times 10^{-44} \text{ s}$
Dp	Planck-scale density	$4.5691019728 \times 10^{96}$	—
Mps	Mahad Prakriti (Planck mass)	$2.203694454 \times 10^{-8}$	$2.176434 \times 10^{-8} \text{ kg}$

A.3 Particle masses (derived)

Symbol	Name	Derived value	Modern equivalent
Pn	Neutron mass	$1.674927646 \times 10^{-27}$	$1.67492750 \times 10^{-27} \text{ kg}$
Pm	Proton mass	$1.672621512 \times 10^{-27}$	$1.67262192 \times 10^{-27} \text{ kg}$
PM	Prakriti Saptha (coherent nuclear mass)	$1.674423179 \times 10^{-27}$	— (not in modern physics)
Mee	Electron mass (measured)	$9.109383824 \times 10^{-31}$	$9.1093837 \times 10^{-31} \text{ kg}$
Me	Mahad Vikrithi (Planck-derived electron)	$9.110233722 \times 10^{-31}$	—

A.4 Electromagnetic & atomic constants

Symbol	Name	Derived value	Modern equivalent
α^{-1}	Fine-structure constant (inverse)	137.04339643	137.035999
kB	Boltzmann constant (= $11/(2 \cdot 7 \cdot C \cdot Mly)$)	1.380658×10^{-23}	1.380649×10^{-23} J/K
e0	Permittivity of free space (Ce)	$8.8469724203 \times 10^{-12}$	$8.8541878 \times 10^{-12}$ F/m
u0	Permeability of free space ($1/(Ce \cdot C^2)$)	$1.2576619439 \times 10^{-6}$	$1.25663706 \times 10^{-6}$ H/m
Si	Impedance of free space	377.03756598	376.730313 Ω
λ_{Cp}	Proton Compton wavelength	$1.3227536624 \times 10^{-15}$ m	$1.32140986 \times 10^{-15}$ m
Ls	Lamb shift	1.05701926×10^9	$\approx 1.057 \times 10^9$ Hz

A.5 Thermodynamic chain constants

Symbol	Name	Derived value	Note
Px	Cyclic pressure exponent	20.947986098	Feeds Boltzmann/temperature chain
KV	Cyclic-volume constant	56.94762837	Feeds Bt
Bt	Base thermal ratio (= $KV \cdot Tk / Px$)	2.7185252132	$\approx e$ (Euler's number)
Tk	Base temperature constant	273.25968858	Reference temperature
Ge	Gyromagnetic/electron derivation constant	0.66478976082	Fills the value left blank in the SoS2 symbols table (resolved via SoS1)

Sourcing note: SoS2's symbols table leaves some derived-constant values blank or quotes measured figures; all values above are reconciled against SoS1 equations. See Registry v1.0 §3 for equation-by-equation provenance.

CHANGELOG

v2.4 — July 16, 2026 (Canonical-source & Registry alignment)

Aligned the guide to the newly locked canonical sources and the master registry:

- **Canonical sources locked.** Header now cites SoS1 (2021-04-21, 189 pp.) and SoS2 (571 pp.), both stored in-repo under `sources/`, resolving the edition/page-drift issue (GAP #1).
- **§2.8a added — "The Two Light Speeds: C (Cs) vs c (CL)."** Reconciles the axiomatic count-rate $C = Cs = 296,575,967$ with the measured $c = CL = 299,792,458$ via the blue-shift factor $Fc \approx 1.010845$ ($CL = Cs \times Fc$), and documents the SoS2 internal C-value inconsistency across pp. 198/295/514/567 (GAP: spectral shift / Fc).
- **§2.8 C row** annotated as $C = Cs$ with a pointer to §2.8a.
- **Appendix A added — "Extended Derived Constants (Vetted)."** New tables of Planck-scale, particle-mass, electromagnetic/atomic, and thermodynamic constants derived from x and C , cross-checked against SoS1 (GAP: derived EM/atomic constants).
- **Registry cross-reference.** The Master Symbol & Terminology Registry v1.0 is named as the authority for symbols, values, and conventions; companion documents (Glossary v2.3, Teacher Guide QA Checklist v1.0) listed in the status block.

v2.3 — June 28, 2026 (14 Technical Corrections + Symbol Rendering Fix)

The following corrections were applied in v2.3, aligning terminology and content to G. Srinivasan's authentic usage. Fixed mathematical symbol rendering issues (exponents and π) in PDF/DOCX output.

#	Section	Change	Rationale
1	§1.2	Changed "Unity becomes duality" → "Natural number counting logic"; added paragraph on scale-invariance of 1, $1/2$ as maximum change increment, and $1/n$ as progressively smaller rates	The axiom's significance is counting logic, not philosophical duality
2	§2.2	Replaced "lean" (describing gradient/rate) with "relative change ratio", "differential rate"	"Lean" is colloquial; authentic Sankhya terminology uses gradient/rate terms
3	§2.3	Rewrote section relative to x (golden mean conjugate), not ϕ ; added x vs. ϕ distinction table	$x = 0.618034$ is the primary Sankhya

#	Section	Change	Rationale
			variable; $\varphi = 1/x$ is derived
4	§2.5	Replaced ✕ / ✓ symbols with plain-text alternatives	Rendering errors on some platforms; replaced with Avoid reducing to: / Technical meaning:
5	§2.6, §2.8, §9.1, §12	Fixed throughout: "golden mean increment" → "golden mean conjugate"; "golden reciprocal" → "golden mean" for φ	$x = 0.618034$ is the conjugate ($1/\varphi$); $\varphi = 1.618034$ is the golden mean
6	All sections	Removed all [bracketed change notes] ([NEW in v2.1 — implements REC-], ← Expanded in v2.1, (v2.1), (v2.2)*) from body text; moved to this Changelog	Body text should read cleanly; version tracking belongs in Changelog
7	§3.0a	Renamed §3.0a → §3.1 (The Detection Principle); shifted old §3.1–§3.8 → §3.2–§3.9	"3.0a" is a non-standard alphanumeric section label
8	§3.2.3	Replaced Atho "Dynamic Asymmetry" content; removed "lean"; described as pre-polarized, latent, undifferentiated, pre-manifestation, pre-radiation state	Atho precedes polarization — "lean" implies directional force which is incorrect at this pre-differentiated stage
9	§3.2.4	Added note on 2×3 Guna matrix for Atho state: Sathwa=0, only Thaama+Raja; balance equation $(\pi \times x)/e = (7 \times RS)/10$; no observable mass/charge/radiation	Explicit matrix description of the Atho state needed for technical completeness
10	§4.5	Added $Kx/C^3 = Mly \times C^3$ equation; explained C^3 symmetry connecting maximum (Kx) and minimum (Mly) through hadronic/leptonic boundary	Shows connection between Kx constant and leptonic mass domain
11	§4.11	Completely rewrote dark matter/energy explanation: "Dark" = $\sim 10^{18}$ orders of Mly below h; $Mly \times C^2 = h$; Dark Matter = hadronic unobservable below h; Dark Energy = energy equivalent of h in Mly counts; removed inaccurate "72% mystery substance" framing	The 72%/28% observational split is a rough analogy; Sankhya explanation is the precise 10^{18} -order h/Mly ratio
12	§10.4	Replaced "The Still Pond (Siddhi)" with "The Porous Pot Immersed in Water (Siddhi)"; added direct quote from SoS2, p. 449	G. Srinivasan's own analogy from the source text; more precise than the "still pond" substitute

#	Section	Change	Rationale
13	§10.5	Rewrote "The Building Floors (C-Power Ladder)" → "The Moolaprakriti Count Ladder (Coherent Simultaneity)"; corrected to describe simultaneous Mly counts from single Mly (bottom) to maximum Kx (top)	The ladder is about simultaneous Moolaprakriti counts, not abstract "C-power floors"
14	Header/ Footer	Incremented version to v2.3; date June 27, 2026; added this Changelog section	Version tracking

v2.2 — June 27, 2026

#	Section	Change	Rationale
1	Throughout	"Product Ratio" → "PHO Ratio" (PR)	More precise terminology aligned to SoS source
2	§4.4.2, §8.2, §12	Added PR = Gmr (Gyromagnetic Ratio) clarification	$PR\ 100/28 = 3.571428$ is identical to the nuclear gyromagnetic ratio measured in physics

v2.1 — June 26, 2026

17 recommendations implemented (REC-3A through REC-3F, REC-4A through REC-4K): - REC-3A: §3.0 Why Sutra 1 Is the Master-Key (new) - REC-3B: §3.1.1–3.1.4 Aikaantha, Aathyantha, Atho, Abhaavath — four properties detailed (new) - REC-3C: §3.6 The Logic of Detection — full step-by-step chain (new) - REC-3D: §3.0a The Detection Principle (new) - REC-3E: §3.4.1 The Guna Equation S.1 — interactive count formulae (new) - REC-3F: §3.8 Scale Invariance of the Substratum (new) - REC-4A: §4.3 The Derivation Chain EP1–EP6 (new) - REC-4B: §4.5 The C-Power Ladder (new) - REC-4C: §4.6 Hadronic States — proton/neutron/PM triad (new) - REC-4D: §4.7 Leptonic States — electron as boundary (new) - REC-4E: §4.9 Planck's Constant axiomatically derived (new) - REC-4F: §4.4.1–4.4.2 RS as power series; PR as nuclear gyromagnetic ratio (new) - REC-4G: §4.10 Wave states, impedance, photon (new) - REC-4H: §4.13 Ferromagnetism as PHO state (new) - REC-4I: §4.11.1 EP8 value (now part of §4.11) - REC-4J: §4.8 Planck mass as PHO oscillator (new) - REC-4K: §4.14 Three PHO papers compared (new)

v2.0 — June 24, 2026

Category 7 analysis reports integrated; consolidated from multiple working documents.

v1.0 — June 2026

Initial release.